Hans-Peter Marshall

Cryosphere Geophysics And Remote Sensing (CryoGARS) group
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Education

2005	Ph.D, Civil Engineering (Geotechnical), GPA: 3.9/4.0 Institute of Arctic and Alpine Research, University of Colorado at Boulder Thesis: *Snowpack spatial variability: towards understanding its effect on remote sensing measurements and snow slope stability*
1999	B.S., Physics with honors, Geophysics Minor (1st in UW history), GPA: 3.6/4.0 University of Washington Thesis: *Wet snow densification*
1994	Nathan Hale High School, Seattle, Washington, GPA: 3.9/4.0

Professional Experience

08/2013-present	Associate Professor, Dept. of Geosciences, Director of CryoGARS, Boise State University
06/2021-present	Leadership team member, NASA SnowEx Mission
10/2021-present	Avalanche Instructor and Senior Advisor, Avalanche Science LLC
06/2018-08/2021	Project Scientist, NASA SnowEx Mission (20+ organizations, 100+ participants)
08/2016-08/2022	Senior Project Staff Engineer, ATA Aerospace, LLC
07/2007-03/2020	Geophysical Engineer (Expert Consultant), U.S. Army Corps of Engineers Cold Regions Research and Engineering Lab (CRREL), Hanover, New Hampshire
08/2008-07/2013	Assistant Professor, Department of Geosciences and Center for Geophysical Investigation of the Shallow Subsurface (CGISS), Boise State University
01/2005-08/2011	Chair of Research Committee, American Avalanche Association
04/2006-08/2008	Research Associate, Inst. of Arctic and Alpine Research (INSTAAR) University of Colorado at Boulder
12/2004-04/2006	Research Geophysical Engineer (GS-11), Student Career Experiences Program (SCEP) U.S. Army Corps of Engineers CRREL, Hanover, New Hampshire & Fairbanks, Alaska
01/2004-07/2004	Invited visiting scientist, Swiss Federal Institute for Snow and Avalanche Research (SLF), Davos, Switzerland
07/2000-12/2005	Research Assistant, INSTAAR, University of Colorado at Boulder
11/2002-12/2003	Avalanche Instructor, Alpine World Ascents, Boulder, Colorado
06/1995-07/2000	Undergraduate Research Assistant, Glaciology Group, University of Washington
08/1999-03/2000	Volunteer high school teacher, Old Crow, Yukon Terr., Canada
Summers 1993-94	Roofer, Seattle, WA

Research Interests

Snow science and glaciology, using geophysics, engineering, and remote sensing
Spatial variability of snow, snow and ice mechanics, snow slope stability, meltwater dynamics

Computational statistics, geostatistics, compressed sensing, instrumentation development

Honors and Awards

2020	BSU Faculty Excellence Award
2019	NASA Instrument Incubator Program Award, NASA, 1st in Idaho history
2019	2018 WRR Editors' Choice Award: Hedrick et al., 2018
2018	NASA Robert H. Goddard Team Award: SnowEx Year 1 Organizing Team
2017	Incoming Chair of the International Snow Working Group for Remote sensing (ISWGR)
2012	Timeless Award, University of Washington, 1 of 150 College of Arts and Sciences alumni awarded from UW history on 150th yr anniv. (1861-2011), 1 of 3 from Physics Department
2010-13	New Investigator Program Award, NASA, 1st in Idaho history
2010	Young Scientist Award, Cryosphere Section, American Geophysical Union
2009	Top Reviewer Award, Cold Regions Science and Technology (CRST)
2008	U.S. Young Scientist Award, International Union of Radio Science (URSI)
2002-05	N.A.S.A. Earth Systems Science Graduate Fellowship
2003	Backcountry Access Avalanche Education Support Grant
2003	Arctic Research Consortium of the U.S. (ARCUS) Award for Arctic Research Excellence, Hon. Mention
2002	University of Colorado Graduate School Student Travel Grant
2001-02	N.S.F. K-12 Engineering Outreach Fellowship
2002	Beverly Sears Graduate Research Grant (University of Colorado)
2001	American Avalanche Association Graduate Research Grant
2000	Li Tze Fong Memorial Fellowship, University of British Columbia (declined)
1999	University of Washington Bonderman Honors Travel Fellowship
1999	Volunteer of the Year, Arctic Canada, Frontiers Foundation
1997-99	University of Washington Mary Gates Undergraduate Research Grant
1999	Featured speaker, University of Washington Undergraduate Research Symposium
1995-98	N.A.S.A. Space Grant Undergraduate Research Program
1999,2001	N.S.F. Graduate Fellowship Program - Honorable Mention
1994-95,97-99	Dean's List, Univerity of Washington
1994-95	U.W. Undergraduate Scholar Award (1994-95)
1994	Washington Athletic Club Student Athlete of the Year, Nathan Hale High School

Student Awards

2018-2020	Joel Gongora, Ph.D. student *NASA Earth Systems Science Fellowship*
2018	Tate Meehan, Ph.D. student *Idaho Space Grant Consortium, NASA EPSCoR graduate fellowship*
2012-2014	Scott Havens, Ph.D. student *NASA Earth Systems Science Fellowship*
2012	Andrew Hedrick, M.S. student *American Avalanche Association Graduate Student Grant*
2012	Scott Havens, Ph.D. student 1^{st} *place poster NSF/NASA EPSCoR annual meeting*
2012	Kris Burch, Lloyd Lowe II, Kolton Drake, Undergrads, *Best Senior Project, BSU College of Engineering*
2003	Adam Maerz, High School student *Colorado State Science Fair Finalist*

Refereed Publications [peer-reviewed publications=70, h-index=28, i10-index=53, citations=2413]

2022 Lievens, H., Brangers, I., **H.-P. Marshall**, Jonas, T., Olefs, M., and DeLannoy, G. (2022). Sentinel-1 snow depth retrieval at sub-kilometer resolution over the European Alps. *The Cryosphere*, 16(1):159–177, doi.org/10.5194/tc–16–159–2022

McGrath, D., Bonnell, R., Zeller, L., Olsen-Mikitowicz, A., Bump, E., Webb, R., and **H.P. Marshall** (accepted (2022)). A time-series of snow density and snow water equivalent observations derived from the integration of gpr and uav sfm observations. *Frontiers in Remote Sensing*

Liu, J., Enderlin, E. M., **H.-P. Marshall**, and Khalil, A. (accepted (2022)). Synchronous retreat of southeast greenland's peripheral glaciers. *Geophysical Research Letters*

Tsang, L., Durand, M., Derksen, C., Barros, A. P., Kang, D.-H., Lievens, H., **H.-P. Marshall**, Zhu, J., Johnson, J., King, J., Lemmetyinen, J., Sandells, M., Rutter, N., Siqueira, P., Nolin, A., Osmanoglu, B., Vuyovich, C., Kim, E. J., Taylor, D., Merkouriadi, I., Brucker, L., Navari, M., Dumont, M., Kelly, R., Kim, R. S., Liao, T.-H., and Xu, X. (accepted (2022)). Review article: Global monitoring of snow water equivalent using high frequency radar remote sensing. *The Cryosphere*

2021 **H.-P. Marshall**, Deeb, E., Forster, R., Vuyovich, C., Elder, K., Hiemstra, C., and Lund, J. (2021). L-band InSAR depth retrieval during the NASA SnowEx 2020 Campaign: Grand Mesa, Colorado. *Proceedings of the International Geoscience and Remote Sensing Symposium (IGARSS)*, pages 625–627, DOI:10.1109/IGARSS47720.2021.9553852

Lewis, G., Osterberg, E., Hawley, B., **H.-P. Marshall**, Meehan, T., Graeter, K. A., McCarthy, F., Overly, T., Thundercloud, Z., Ferris, D., Koffman, B., and Dibb, J. (2021). Atmospheric blocking drives recent albedo change across the western Greenland ice sheet percolation zone. *Geophysical Research Letters*, 48(10):e2021GL092814

Johnson, J., Anderson, J., **H.-P. Marshall**, Havens, S., and Watson, L. (2021). Snow avalanche detection and source constraints made using multiple infrasound sensor arrays. *Journal of Geophysical Research-Earth Surface*, 126(3):e2020JF005741

Liu, J., Enderlin, E. M., **H.P. Marshall**, and Khalil, A. (2021). Automated detection of marine glacier calving fronts using the 2d wavelet transform modulus maxima (wtmm) segmentation method. *IEEE Transactions On Geoscience and Remote Sensing*, 59(11):9047–9056, doi:10.1109/TGRS.2021.3053235

Meehan, T. G., **H.P. Marshall**, Bradford, J. H., Hawley, R. L., Overly, T. B., Lewis, G., Graeter, K., Osterburg, E., and McCarthy, F. (2021). Historical surface mass balance reconstruction 1984-2017 from greentracs multi-offset ground-penetrating radar. *Journal of Glaciology*, 67(262):219–228, doi:10.1017/jog.2020.91

Hojatimalekshah, A., Uhlmann, Z., Glenn, N. F., Hiemstra, C. A., Tennant, C. J., Graham, J. D., Spaete, L., Gelvin, A., **H.P. Marshall**, McNamara, J., and Enterkine, J. (2021). Tree canopy and snow depth relationships at fine scales with terrestrial laser scanning. *The Cryosphere*, 15(5):2187–2209, doi:10.5194/tc–15–2187–2021

Kim, R. S., Kumar, S., Vuyovich, C., Houser, P., Lundquist, J., Mudryk, L., Durand, M., Barros, A., Kim, E. J., Forman, B. A., Gutmann, E. D., Wrzesien, M. L., Garnaud, C., Sandells, M., **H.P. Marshall**, Cristea, N., Pflug, J. M., Johnston, J., Cao, Y., Mocko, D., and Wang, S. (2021). Snow Ensemble Uncertainty Project (SEUP): Quantification of snow water equivalent uncertainty across north america via ensemble land surface modeling. *The Cryosphere*, 15:771–791, doi:10.5194/tc–15–771–2021

Webb, R. W., Marziliano, A., McGrath, D., Bonnell, R., Meehan, T. G., Vuyovich, C., and **H.-P. Marshall** (2021). In situ determination of dry and wet snow permittivity: Improving equations for low frequency radar applications. *Remote Sensing*, 13:4617, DOI:10.3390/rs13224617

Bonnell, R., McGrath, D., Williams, K., Webb, R., Fassnacht, S. R., and **H.-P. Marshall** (2021). Spatiotemporal variations in liquid water content in a seasonal snowpack: Implications for radar remote sensing. *Remote Sensing*, 13:4223, DOI:10.3390/rs13214223

Lievens, Hans and Brangers, Isis and Marshall, Hans-Peter and Jonas, Tobias and Olefs, Marc and De Lannoy, Gabriëlle (2021). Observing snow depth at sub-kilometer resolution over the european alps from sentinel-1. *2021 IEEE International Geoscience and Remote Sensing Symposium IGARSS*, pages 618–621, doi:10.1109/IGARSS47720.2021.9553420

Tsang, L., Durand, M., Derksen, C., Barros, A. P., Kang, D.-H., Lievens, H., **Marshall, H.P.**, Zhu, J., Johnson, J., King, J., Lemmetyinen, J., Sandells, M., Rutter, N., Siqueira, P., Nolin, A., Osmanoglu, B., Vuyovich, C., Kim, E. J., Taylor, D., Merkouriadi, I., Brucker, L., Navari, M., Dumont, M., Kelly, R., Kim, R. S., Liao, T.-H., and Xu, X. (2021). Review article: Global monitoring of snow water equivalent using high frequency radar remote sensing. *The Cryosphere Discussions*, pages doi.org/10.5194/tc–2021–295

2020 Hedrick, A. R., Marks, D., **H.P. Marshall**, McNamara, J., Havens, S., Trujillo, E., Sandusky, M., Robertson, M., Johnson, M., Bormann, K. J., and Painter, T. H. (2020). From drought to flood: A water balance analysis of the tuolumne river basin during extreme conditions (2015 – 2017). *Hydrol. Process.*, 34(11):2560–2574

Webb, R., Raleigh, M., McGrath, D., Molotch, N., Elder, K., Hiemstra, C., Brucker, L., and **H.P. Marshall** (2020). Within-stand boundary effects of snow water equivalent distribution in forested areas. *Water Resources Research*, 56:e2019WR024905

Lund, J., Forster, R. R., Rupper, S. B., **H.P. Marshall**, Deeb, E. J., and Hashmi, M. (2020). Mapping snowmelt progression in the upper indus basin with synthetic aperture radar. *Frontiers in Earth Science*, 7:318, doi:10.3389/feart.2019.00318

2019 McGrath, D., Webb, R., Shean, D., Bonnell, R., **H.P. Marshall**, Painter, T., Molotch, N., Elder, K., Hiemstra, C., and Brucker, L. (2019). Spatially extensive ground-penetrating radar snow depth observations during NASA's 2017 SnowEx Campaign: Comparison to in situ, airborne, and satellite observations. *Water Resources Research*, 55(11):10026–10036, doi:10.1029/2019WR024907

Lievens, H., Demuzere, M., **H.P. Marshall**, Reichle, R. H., Brucker, L., Brangers, I., de Rosnay, P., Dumont, M., Girotto, M., Immerzeel, W. W., Jonas, T., Kim, E. J., Koch, I., Marty, C., Saloranta, T., Schöber, J., and Lannoy, G. J. D. (2019). Snow depth variability in the northern hemisphere mountains observed from space. *Nature Communications*, 10(1):1–12

Lewis, G., Osterberg, E., Hawley, R., Marshall, H. P., Meehan, T., Graeter, K., McCarthy, F., Overly, T., Thundercloud, Z., and Ferris, D. (2019a). Recent precipitation decrease across the western greenland ice sheet percolation zone. *The Cryosphere*, 13:2797–2815, doi:10.5194/tc–13–2797–2019

2018 Graeter, K. A., Osterberg, E. C., Ferris, D. G., Hawley, R. L., **H.P. Marshall**, Lewis, G., Meehan, T., McCarthy, F., and Birkel, S. D. (2018). Ice core records of West Greenland surface melt and climate forcing. *Geophysical Research Letters*, 45(7):3164–3172, doi:10.1002/2017GL076641

Hedrick, A., Marks, D., Kormos, P., **H.P. Marshall**, Havens, S., Bormann, K., and Painter, T. H. (2018). Direct insertion of NASA Airborne Snow Observatory-derived snow depth time-series into the iSnobal energy balance snow model. *Water Resources Research*, 54(10):8045–8063, doi.org/10.1029/2018WR023190

McGrath, D., Sass, L., O'Neel, S., McNeil, C., Candela, S. G., Baker, E. H., and **H.P. Marshall** (2018). Interannual snow accumulation variability on glaciers derived from repeat, spatially extensive ground-penetrating radar surveys. *The Cryosphere*, 12:3617–3633, doi.org/10.5194/tc–12–3617–2018

Hagenmuller, P., van Herwijnen, A., Pielmeier, C., and **H.P. Marshall** (2018). Evaluation of the snow penetrometer Avatech SP2. *Cold Regions Science and Technology*, 149:83–94

McNamara, J. P., Benner, S. G., Poulos, M. J., Pierce, J. L., Chandler, D. G., Kormos, P. R., **H.P. Marshall**, Flores, A. N., Seyfried, M., Glenn, N. F., and Aishlin, P. (2018). Form and function relationships revealed by long-term research in a semiarid mountain catchment. *WIREs Water*, 5(2):e1267, doi:10.1002/wat2.1267

Brucker, L., Hiemstra, C., Marshall, H.-P., Elder, K., De Roo, R., Mousavi, M., Bliven, F., Peterson, W., Deems, J., Gadomski, P., Gelvin, A., Spaete, L., Barnhart, T., Brandt, T., Burkhart, J., Crawford, C., Datta, T., Erikstrod, H., Glenn, N., Hale, K., Holben, B., Houser, P., Jennings, K., Kelly, R., Kraft, J., Langlois, A., McGrath, D., Merriman, C., Molotch, N., Nolin, A., Polashenski, C., Raleigh, M., Rittger, K., Rodriguez, C., Roy, A., Skiles, M., Small, E., Tedesco, M., Tennant, C., Thompson, A., Uhlmann, Z., Webb, R., and Wingo, M. (2018a). Nasa SnowEx'17 in situ Measurements and Ground-Based Remote Sensing. *IGARSS 2018 - 2018 IEEE International Geoscience and Remote Sensing Symposium*, pages 6266–6268, doi:10.1109/IGARSS.2018.8517777

Kim, Y., Reck, T. J., Alonso-delPino, M., Painter, T. H., **H.P. Marshall**, Bair, E. H., Dozier, J., Chattopadhyay, G., Liou, K.-N., Chang, M.-C. F., and Tang, A. (2018). A Ku-band CMOS FMCW radar transceiver for snowpack remote sensing. *IEEE Transactions On Microwave Theory and Techniques*, 66(5):2480–2494

2017 Mikesell, T., van Wijk, K., Otheim, L., **H.P. Marshall**, and Kurbatov, A. (2017). Laser ultrasound observations of mechanical property variations in ice cores. *Geosciences*, 7(3):47

Lewis, G., Osterberg, E., Hawley, R., Whitmore, B., **H.P. Marshall**, and Box, J. (2017a). Regional Greenland accumulation variability from Operation IceBridge airborne accumulation radar. *The Cryosphere*, 11(2):773

Deeb, E. J., Marshall, H.-P., Forster, R. R., Jones, C. E., Hiemstra, C. A., and Siqueira, P. R. (2017). Supporting NASA SnowEx remote sensing strategies and requirements for L-band interferometric snow depth and snow water equivalent estimation. *2017 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, pages 1395–1396, doi:10.1109/IGARSS.2017.8127224

Kim, E., Gatebe, C., Hall, D., Newlin, J., Misakonis, A., Elder, K., Marshall, H.-P., Hiemstra, C., Brucker, L., De Marco, E., Crawford, C., Kang, D. H., and Entin, J. (2017). NASA's SnowEx campaign: Observing seasonal snow in a forested environment. *2017 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, pages 1388–1390, doi:10.1109/IGARSS.2017.8127222

Pegau, W. S., Garron, J., Zabilansky, L., Bassett, C., Bello, J., Bradford, J., Carns, R., Courville, Z., Eicken, H., Elder, B., Eriksen, P., Lavery, A., Light, B., Maksym, T., **H.P. Marshall**, Oggier, M., Perovich, D., Pacwiardowski, P., Singh, H., Tang, D., Wiggins, C., and Wilkinson, J. (2017). Detection of oil in and under ice. *International Oil Spill Conference Proceedings*, 2017(1):1857–1876

Brucker, L., Hiemstra, C., Marshall, H.-P., Elder, K., De Roo, R., Mousavi, M., Bliven, F., Peterson, W., Deems, J., Gadomski, P., Gelvin, A., Spaete, L., Barnhart, T., Brandt, T., Burkhart, J., Crawford, C., Datta, T., Erikstrod, H., Glenn, N., Hale, K., Holben, B., Houser, P., Jennings, K., Kelly, R., Kraft, J., Langlois, A., McGrath, D., Merriman, C., Molotch, N., Nolin, A., Polashenski, C., Raleigh, M., Rittger, K., Rodriguez, C., Roy, A., Skiles, M., Small, E., Tedesco, M., Tennant, C., Thompson, A., Tian, L., Uhlmann, Z., Webb, R., and Wingo, M. (2017). A first overview of SnowEx ground-based remote sensing activities during the winter 2016–2017. *2017 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, pages 1391–1394, doi:10.1109/IGARSS.2017.8127223

2016 Evans, S., Flores, A., Heilig, A., Kohn, M., **Marshall, H.P.**, and McNamara, J. (2016). Isotopic evidence for lateral flow and diffusive transport, but not sublimation, in a sloped seasonal snowpack, Idaho, U.S.A. *Geophysical Research Letters*, 43(7):3298–3306

Rutter, N., **Marshall, H.P.**, Tape, K., Essery, R., and King, J. (2016). Impact of spatial averaging on radar reflectivity at internal snowpack layer boundaries. *Journal of Glaciology*, 62(236):1065–1074

Bradford, J. H., Babcock, E. L., **Marshall, H.P.**, and Dickins, D. F. (2016). Targeted reflection-waveform inversion of experimental ground-penetrating radar data for quantification of oil spills under sea ice. *Geophysics*, 81(1):WA59–WA70

2015 Heilig, A., Mitterer, C., Schmid, L., Wever, N., Schweizer, J., **Marshall, H.P.**, and Eisen, O. (2015). Seasonal and diurnal cycles of liquid water in snow-measurements and modeling. *Journal of Geophysical Research-Earth Surface*, 120(10):2139–2154

Kormos, P. R., McNamara, J. P., Seyfried, M. S., **Marshall, H.P.**, Marks, D., and Flores, A. N. (2015). Bedrock infiltration estimates from a catchment water storage-based modeling approach in the rain snow transition zone. *Journal of Hydrology*, 525:231–248

Hedrick, A., **Marshall, H.P.**, Winstral, A., Elder, K., Yueh, S., and Cline, D. (2015). Independent evaluation of the SNODAS snow depth product using regional-scale LiDAR-derived measurements. *Cryosphere*, 9(1):13–23

2014 Kormos, P. R., Marks, D., McNamara, J. P., **Marshall, H.P.**, Winstral, A., and Flores, A. N. (2014a). Snow distribution, melt and surface water inputs to the soil in the mountain rain-snow transition zone. *Journal of Hydrology*, 519:190–204

Walters, R. D., Watson, K. A., **Marshall, H.P.**, McNamara, J. P., and Flores, A. N. (2014). A physiographic approach to downscaling fractional snow cover data in mountainous regions. *Remote Sensing of Environment*, 152:413–425

Havens, S., **Marshall, H.P.**, Johnson, J. B., and Nicholson, B. (2014b). Calculating the velocity of a fast-moving snow avalanche using an infrasound array. *Geophysical Research Letters*, 41(17):6191–6198

Anderson, B. T., McNamara, J. P., **Marshall, H.P.**, and Flores, A. N. (2014). Insights into the physical processes controlling correlations between snow distribution and terrain properties. *Water Resources Research*, 50(6):4545–4563

Tinkham, W. T., Smith, A. M. S., **Marshall, H.P.**, Link, T. E., Falkowski, M. J., and Winstral, A. H. (2014). Quantifying spatial distribution of snow depth errors from lidar using random forest. *Remote Sensing of Environment*, 141:105–115

McCreight, J. L., Slater, A. G., **Marshall, H.P.**, and Rajagopalan, B. (2014). Inference and uncertainty of snow depth spatial distribution at the kilometre scale in the colorado rocky mountains: the effects of sample size, random sampling, predictor quality, and validation procedures. *Hydrological Processes*, 28(3):933–957

Kormos, P. R., Marks, D., Williams, C. J., **Marshall, H.P.**, Aishlin, P., Chandler, D. G., and McNamara, J. P. (2014b). Soil, snow, weather, and sub-surface storage data from a mountain catchment in the rain-snow transition zone. *Earth System Science Data*, 6(1):165–173

2013 Havens, S., **Marshall, H.P.**, Pielmeier, C., and Elder, K. (2013b). Automatic grain type classification of snow micro penetrometer signals with random forests. *IEEE Transactions On Geoscience and Remote Sensing*, 51(6):3328–3335

Eiriksson, D., Whitson, M., Luce, C. H., **Marshall, H.P.**, Bradford, J., Benner, S. G., Black, T., Hetrick, H., and McNamara, J. P. (2013). An evaluation of the hydrologic relevance of lateral flow in snow at hillslope and catchment scales. *Hydrological Processes*, 27(5):640–654

Tinkham, W. T., Hoffman, C. M., Falkowski, M. J., Smith, A., **Marshall, H.P.**, and Link, T. E. (2013a). A methodology to characterize vertical accuracies in lidar-derived products at landscape scales. *Photogrammetric Engineering & Remote Sensing*, 79(8):709–716

Rott, H., Nagler, T., Prinz, R., Voglmeier, K., Fromm, R., Adams, M. S., Yueh, S., Elder, K., **Marshall, H.P.**, Coccia, A., Imbembo, E., Schuttemeyer, D., and Kern, M. (2013). AlpSAR 2012-13, a field experiment on snow observations and parameter retrievals with Ku-and X-band radar. *ESA Living Planet Symposium*, ESA SP-722.2-13:92

2012 Mikesell, T. D., van Wijk, K., Haney, M. M., Bradford, J. H., **Marshall, H.P.**, and Harper, J. T. (2012). Monitoring glacier surface seismicity in time and space using rayleigh waves. *Journal of Geophysical Research-Earth Surface*, 117(F2):F02020

2011 **Marshall, H.P.**, Schweizer, J., and Birkeland, K. (2011). Some recent advances in snow and avalanche science. *Cold Regions Science and Technology*, 69(2-3):119–121

Geroy, I., Gribb, M., **Marshall, H.P.**, Chandler, D., Benner, S. G., and McNamara, J. P. (2011). Aspect influences on soil water retention and storage. *Hydrological Processes*, 25(25):3836–3842

2010 Tape, K. D., Rutter, N., **Marshall, H.P.**, Essery, R., and Sturm, M. (2010b). Recording microscale variations in snowpack layering using near-infrared photography. *Journal of Glaciology*, 56(195):75–80

Tape, K. D., Lord, R., **Marshall, H.P.**, and Ruess, R. W. (2010a). Snow-mediated ptarmigan browsing and shrub expansion in arctic alaska. *Ecoscience*, 17(2):186–193

Koh, G., Lever, J., Arcone, S., **Marshall, H.P.**, and Ray, L. (2010). Autonomous fmcw radar survey of antarctic shear zone. *IEEE Publications, 13th International Conference on Ground Penetrating Radar*, DOI: 10.1109/ICGPR.2010.5550174

2009 **Marshall, H.P.** and Johnson, J. B. (2009). Accurate inversion of high-resolution snow penetrometer signals for microstructural and micromechanical properties. *Journal of Geophysical Research-Earth Surface*, 114(F04016)

Lutz, E., Birkeland, K., and **Marshall, H.P.** (2009b). Quantifying changes in weak layer microstructure associated with artificial load changes. *Cold Regions Science and Technology*, 59(2-3):202–209

Pielmeier, C. and **Marshall, H.P.** (2009). Rutschblock-scale snowpack stability derived from multiple quality-controlled snowmicropen measurements. *Cold Regions Science and Technology*, 59(2-3):178–184

2008 **Marshall, H.P.** and Koh, G. (2008). FMCW radars for snow research. *Cold Regions Science and Technology*, 52(2):118–131

Hardy, J., Davis, R., Koh, Y., Cline, D., Elder, K., Armstrong, R., **Marshall, H.P.**, Painter, T., Saint-Martin, G. C., Deroo, R., Sarabandi, K., Graf, T., Koike, T., and McDonald, K. (2008). NASA Cold Land Processes Experiment (CLPX 2002/03): Local Scale Observation Site. *Journal of Hydrometeorology*, 9(6):1434–1442

H.-P. Marshall, Koh, G., and Sturm, M. (2008b). Ultra-broadband portable microwave fmcw radars for measuring snow depth, snow water equivalent, and stratigraphy: practical considerations. *Proceedings of the 29th International Union of Radio Science (URSI) General Assembly*, page DOI:10.1.1.498.5307. (Invited)

2007 **Marshall, H.P.**, Schneebeli, M., and Koh, G. (2007). Snow stratigraphy measurements with high-frequency FMCW radar: Comparison with snow micro-penetrometer. *Cold Regions Science and Technology*, 47(1-2):108–117

O'Neel, S., **Marshall, H.P.**, McNamara, D. E., and Pfeffer, W. T. (2007). Seismic detection and analysis of icequakes at Columbia Glacier, Alaska. *Journal of Geophysical Research-Earth Surface*, 112(F3)

Waddington, E. D., Neumann, T. A., Koutnik, M. R., **Marshall, H.P.**, and Morse, D. L. (2007). Inference of accumulation-rate patterns from deep layers in glaciers and ice sheets. *Journal of Glaciology*, 53(183):694–712

2005 **Marshall, H.P.**, Koh, G., and Forster, R. R. (2005). Estimating alpine snowpack properties using FMCW radar. *Annals of Glaciology*, 40:157–162

2004 **Marshall, H.P.**, Koh, G., and Forster, R. R. (2004b). Ground-based frequency-modulated continuous wave radar measurements in wet and dry snowpacks, Colorado, USA: an analysis and summary of the 2002-03 NASA CLPX data. *Hydrological Processes*, 18(18):3609–3622

2002 **Marshall, H.P.**, Harper, J. T., Pfeffer, W. T., and Humphrey, N. F. (2002). Depth-varying constitutive properties observed in an isothermal glacier. *Geophysical Research Letters*, 29(23):61–1:61–4

1999 **Marshall, H.P.**, Conway, H., and Rasmussen, L. A. (1999). Snow densification during rain. *Cold Regions Science and Technology*, 30(1-3):35–41

Conway, H., Rasmussen, L. A., and **Marshall, H.P.** (1999). Annual mass balance of Blue Glacier, USA: 1955-97. *Geografiska Annaler, Ser. A-Phys. Geogr.*, 81A(4):509–520

Morse, D. L., Waddington, E. D., **Marshall, H.P.**, Neumann, T. A., Steig, E. J., Dibb, J. E., Winebrenner, D. P., and Arthern, R. J. (1999). Accumulation rate measurements at Taylor Dome, East Antarctica: Techniques and strategies for mass balance measurements in polar environments. *Geografiska Annaler, Ser. A-Phys. Geogr.*, 81A(4):683–694

Refereed Book Chapters / Technical Reports

2021 Arenson, L., Colgan, W., and **Marshall, H.P.** (2021, 2nd edition). *Physical, Thermal, and Mechanical Properties of Snow, Ice, and Permafrost*, chapter 2, pages 35–75. Snow and ice-related hazards, risks, and disasters. Elsevier, ed. Haeberli and Whiteman, Radarweg 29, PO Box 211, 1000 AE Amsterdam, Netherlands

Youngblood, P. and **H.-P. Marshall** (2021). Continuous snow temperature monitoring for the idaho highway 21 avalanche program. Technical Report FHWA-ID-21-276, Idaho Transportation Department, 3311 W State Street, P. O. Box 7129, Boise, ID 83707-1129 United States

2019 Meehan, T. G., **H.-P. Marshall**, Deeb, E. J., and Shoop, S. A. (2019). Snowmicropenetrometer applications for winter vehicle mobility. ERDC 6.2 Boreal Aspects of Ensured Maneuver (BAEM) TR-19-14, U.S. Army Cold Regions Research and Engineering Laboratory

2015 **Marshall, H.P.**, Hawley, R., and Tedesco, M. (2015). *Field measurements for remote sensing of the cryosphere*, chapter 14, pages 345–381. Remote Sensing of the Cryosphere. John Wiley & Sons, Ltd, ed. M. Tedesco

Arenson, L., Colgan, W., and **Marshall, H.P.** (2015). *Physical, Thermal, and Mechanical Properties of Snow, Ice, and Permafrost*, chapter 2, pages 35–75. Snow and ice-related hazards, risks, and disasters. Elsevier, ed. Haeberli and Whiteman, Radarweg 29, PO Box 211, 1000 AE Amsterdam, Netherlands

2014 Havens, S., **Marshall, H.P.**, Trisca, G., and Johnson, J. B. (2014c). Real time avalanche detection for high risk areas. Monograph FHWA-ID-14-219, Idaho Transportation Department, 3311 W State Street, P. O. Box 7129, Boise, ID 83707-1129 United States

Peer Reviewed Archived Datasets

2022 Breen, C. M., Lumbrazo, C., Hiemstra, C., Vuyovich, C. M., Raleigh, M. S., and **H.-P. Marshall** (2022). Snowex20 time-lapse imagery, version 1. Digital media, https://doi.org/10.5067/P3D1QRH7O8X5, NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA

McDowell, I., Keegan, K., Osterberg, E., Hawley, R., and **H.P. Marshall** (2022). Grain size and ice layer stratigraphy from the upper 10 meters of cores 1-7 collected during the 2016 greenland traverse for accumulation and climate studies (greentracs), southwestern greenland. Technical report, Arctic Data Center

2021 Brucker, L., Vuyovich, C. M., Elder, K., **H.-P. Marshall**, and Hiemstra, C. (2021). Snowex20 grand mesa autumn 2019 snow pits, version 1. Technical Report doi: https://doi.org/10.5067/4IZGCQC1J31Q, NASA National Snow and Ice Data Center Distributed Active Archive Center

Hiemstra, C., Vuyovich, C. M., and **H.-P. Marshall** (2021). Snowex20 grand mesa reference gis data sets, version 1. Technical Report doi: https://doi.org/10.5067/YDZXY4Q79VIJ, NASA National Snow and Ice Data Center Distributed Active Archive Center

Vuyovich, C. M., **H.-P. Marshall**, Elder, K., Hiemstra, C., Brucker, L., and McCormick, M. (2021). Snowex20 grand mesa intensive observation period snow pit measurements, version 1. Technical Report doi: https://doi.org/10.5067/DUD2VZEVBJ7S, NASA National Snow and Ice Data Center Distributed Active Archive Center

2020 **H.P. Marshall** and Gleason, A. (2020). SnowEx17 Senator Beck SnowMicroPen (SMP) Raw Penetration Force Profiles, Version 1. Technical Report doi:https://doi.org/10.5067/K7P8PB6EO1HO, NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA

Hiemstra, C., **H.-P. Marshall**, Vuyovich, C. M., Elder, K., Mason, M., and Durand, M. (2020). SnowEx20 Community Snow Depth Probe Measurements, Version 1. Technical Report doi: https://doi.org/10.5067/9IA978JIACAR, NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA

Mason, M., **H.-P. Marshall**, and Merkouriadi, I. (2020). SnowEx20 SnowMicroPen (SMP) Raw Penetration Force Profiles, Version 1. Technical Report doi: https://doi.org/10.5067/ZYW6IHFRYDSE, NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA

Johnson, J., Anderson, J., **H.-P. Marshall**, Havens, S., and Watson, L. (2020). Dataset on fortress snow avalanche 13 march 2018. *Boise State University Infrasound Repository*, 4

2018 Elder, K., Brucker, L., Hiemstra, C., and **H.P. Marshall** (2018). Snowex17 community snow pit measurements, version 1. Digital media, https://doi.org/10.5067/Q0310G1XULZS, NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA

2018 Brucker, L., Hiemstra, C., **H.P. Marshall**, and Elder, K. (2018b). Snowex17 community snow depth probe measurements, version 1. Digital media, https://doi.org/10.5067/WKC6VFMT7JTF, NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA

2004 **Marshall, H.P.**, Koh, G., and Forster, R. (2004a). CLPX-Ground: Ground-based Frequency Modulated Continuous Wave (FMCW) radar. Digital media, http://nsidc.org/data/nsidc-0164.html, National Snow and Ice Data Center, University of Colorado at Boulder

Magazine Articles / Podcasts

2013 Durham, T., **Marshall, H.P.**, Tsang, L., Racette, P., Bonds, Q., Miranda, F., and Vanhille, K. (2013). Wideband sensor technologies for measuring surface snow. *Earthzine*, December 2(earthzine.org)

2012 **Marshall, H.P.** (2012). Researching snow: More than just hitting the slopes. Podcast https://beyondtheblue.boisestate.edu/blog/2012/01/09/hp-marshall/, Beyond the Blue: Boise State Faculty Podcasts

2011 Sturm, M. and **Marshall, H.P.** (2011). Marshall receives Cryosphere Young Investigator Award (citation and response). *EOS Transactions, AGU*, 92(22):189

Research Funding Summary

Agency	**Short Title**	**Role**	**Dates**	**Amount**
CRREL	Advancement of snow monitoring	PI	9/2020-8/2022	$1,000,000
NASA IIP	Meta-surface Radar for Snow Remote Sensing	PI	2/2020-1/2023	$2,168,639
U.S. DOTs	Enhanced Infrasound Avalanche Detection	PI	6/2017-8/2018	$99,509
NSF	MRI: Long-range UAV for Sea-Ice Monitoring	Co-PI	9/2018-8/2022	$1,615,385
ID Dept Commerce	A General Purpose Goniometer	Co-PI	6/2018-7/2020	$368,772
NASA	Earth Systems Science Fellowship (Gongora)	PI	09/2018-8/2019	$90,000
ITD	Continuous snow temperature monitoring	PI	9/2018-8/2020	$45,000
U.S. DOTs	Infrasound Avalanche Monitoring	Co-PI	1/2019 - 12/2022	$99,509
NASA	NASA SnowEx Ground-based Radar	Co-PI	6/2018-6/2021	$444,742
NASA	NASA SnowEx Y1 Data Assimilation	Co-PI	6/2018-6/2021	$421,212
NASA	Spatiotemporal Patterns in GBRS	PI	6/2017-6/2022	$460,719
NASA	L-Band InSAR for snow depth and SWE	Co-PI	6/2017-6/2022	$450,000
NASA JPL	ASO and UAVSAR data fusion	PI	6/2017-7/2018	$21,103
U.S. DOTs	Infrasound Avalanche Detection	PI	6/2017-8/2018	$99,339
NASA	Broadband Array IIP - Phase 2	PI	02/2015-05/2017	$330,000
NSF	GreenTRACS	PI	01/2015-12/2017	$355,338
NASA	Multi-frequency Data Assimilation	Co-PI	07/2015-06/2018	$295,577
NASA	Monitoring Earth's Hydrosphere	Co-PI	06/2014-05/2017	$749,870
U.S. Army	SWE estimates using radar phase	Co-PI	10/2013- 9/2016	$490,800
OGP	Remote Sensing Tests with Oil in Ice	Co-PI	01/2014-12/2014	$128,739
NASA	NASA Snow School	Co-PI	01/2014-12/2014	$65,385
NSF	MRI: Laser-ultrasonic Ice Core Tomography	Co-PI	09/2012-08/2013	$336,122
NSF	MRI:GPU Computing/Visualization Cluster	Co-PI	09/2012-08/2013	$555,384
NSF	Measuring Sublimation with Isotopes	Co-PI	04/2012-04/2013	$19,912
ITD	Avalanche Detection	PI	09/2012-08/2015	$100,000
NASA	WISM IIP - Phase 1	PI	11/2011-12/2014	$94,947
NASA	Remote Sensing of the Cryosphere	PI	01/2011-12/2013	$1,124,552
NASA	Improvement of Remotely Sensed SWE	PI	09/2010-08/2013	$329,953
NSF	Lateral Flow of Water in Snow	PI	04/2010-03/2013	$228,963
NASA	Earth Systems Science Fellowship (Havens)	PI	09/2010-12/2013	$90,000
D.F. Dickins	Oil Detection Radar	PI	01/2010-12/2012	$328,747
IDT	Avalanche Modeling	PI	09/2010-08/2012	$10,000
NASA	CLPX-II Grand Mesa, Colorado	PI	08/2010-08/2012	$35,468
NASA	Investigation of Snowpack Properties	PI	08/2006-07/2009	$462,132
NSF	Radar for Measuring Water Pathways	PI	09/2005-08/2010	$69,785
			TOTAL	**$13,585,603**

Awarded Research Grants

9/2020-8/2022 U.S. Army Cold Regions Research and Engineering Laboratory BAA

Advancement of snow monitoring for water resources, vehicle mobility, and hazard mitigation: using optical, microwave, acoustic, and seismic techniques
PI, $1,000,000
Co-PIs: Ellyn Enderlin, Dylan Mikesell, Lee Liberty, Jeff Johnson, Jake Anderson

2/2020-1/2023 NASA Instrument Incubator Program

Digitally Enhanced Meta-surface Radar/Radiometer for Snow Remote Sensing
PI, $2,168,639
Co-PIs: Adrian Tang (JPL), Frank Chang (UCLA)

7/2019-12/2021 Consortium of U.S. Dept. of Transportation Avalanche Forecasting Programs

Infrasound Avalanche Monitoring: Enhanced Monitoring in Little Cottonwood Canyon, Utah
Co-PI, $99,509
PI: Jeff Johnson (BSU)

9/2018-8/2022 NSF: Major Research Instrumentation

Development of a Long-Range UAV For Sea Ice Monitoring
Co-PI, $1,615,385
PI: Andy Mahoney (UAF), CoPIs: Hajo Eiken, Dejan Raskovic, Denise Thorsen, Michael Hatfield (UAF)

6/2018-7/2020 Idaho Department of Commerce

A General Purpose Goniometer
Co-PI, $368,772
PI: Sing Ming Loo (BSU)

09/2018-8/2019 NASA Earth and Space Science Fellowship

Optimal Sampling Strategies for Snow: Application of Machine Learning and Graph Theory
Principal Investigator, $90,000
Student: Joel Gongora

09/2018-12/2020 Idaho Department of Transportation Research Program

Continuous snow temperature monitoring at Banner Summit, Idaho
Principal Investigator, $45,000

1/2018-5/2019 Consortium of U.S. Department of Transportation Avalanche Forecasting Programs

Infrasound Avalanche Monitoring: Enhanced Monitoring in Little Cottonwood Canyon, Utah
Co-PI, $99,509
PI: Jeff Johnson, Co-PI: Sin Ming Loo (BSU)

6/2018-6/2021 NASA Terrestrial Hydrology Program 2017

Constraining airborne and satellite algorithms for estimating snow water equivalent through spatially extensive ground penetrating radar observations during NASA's SnowEx campaign

Co-PI, $444,742
PI: Dan McGrath (Colorado State Univ.), Co-PI: Ryan Webb (Univ. Colorado)

6/2018-6/2021 NASA Terrestrial Hydrology Program 2017

Using SnowEx Year 1 Data to Assess and Improve Snow Measurements, Validation, Algorithms, and Data Assimilation in Forest and Shrubland Landscapes
Co-PI, $421,212
PI: Chris Hiemstra (CRREL), Co-PIs: Kelly Elder (USFS), Ludo Brucker (NASA Goddard)

6/2017-6/2022 NASA Terrestrial Hydrology Program 2016

Spatiotemporal patterns in ground-based snow remote sensing and in-situ observations: Analysis of CLPX-II, CoreH2O, and SnowEx Y1 data for sampling design and site selection
Principal Investigator, $460,719
Co-PIs: Kelly Elder (USFS), Eli Deeb (CRREL), Ludo Brucker (NASA Goddard)

6/2017-6/2022 NASA Terrestrial Hydrology Program 2016

Remote sensing strategies and requirements for L-Band interferometric snow depth and snow water equivalent estimation
Co-PI, $450,000
PI: Eli Deeb (CRREL), Co-PIs: Rick Forster (Univ. Utah), Cathleen Jones (NASA JPL), Chris Hiemstra (CRREL)

6/2017-8/2018 NASA Jet Propulsion Laboratory, CalTech

ASO and UAVSAR Data Fusion: Exploring the Advantages of Combining Airborne LIDAR and InSAR
Principal Investigator, $21,103

6/2017-8/2018 Consortium of U.S. Department of Transportation Avalanche Forecasting Programs

Infrasound Avalanche Detection: Comparison Study of Multiple New Technologies
Principal Investigator, $99,339
Co-PIs: Jeff Johnson, Sin Ming Loo (BSU)

2/2015-5/2017 NASA Instrument Incubator Program (IIP) - Phase 2

Broadband Array Technology in Support of the Snow and Cold Land Process (SCLP) Mission
Principal Investigator, $330,000

1/2015-12/2017 NSF Office of Polar Programs

Collaborative Research: GreenTrACS: Greenland Traverse for Accumulation and Climate Studies
Principal Investigator, $355,338
Co-PI: John Bradford (BSU)

7/2015-6/2018 NASA Terrestrial Hydrology Program (THP)

Multiple Frequency Active Microwave Remote Sensing for Snow Water Equivalent Retrieval from Space: A Data Assimilation Approach
Co-PI, $295,577

PI: Lejo Flores (BSU)

6/2014-6/2018 NASA Experimental Program to Stimulate Competitive Research (EPSCoR)

Monitoring Earth's Hydrosphere: Integrating Remote Sensing, Modeling, and Verification

Co-PI, $749,870

PI: Lejo Flores, Co-PIs: Jim McNamara, Nancy Glen (BSU)

10/2013-9/2016 U.S. Army Basic Research Program

Quantifying the accuracy of Snow Water Equivalent estimates using radar phase over a wide range of frequencies and varying snow conditions

Co-PI, $490,800

PI: Eli Deeb (CRREL), Co-PI: Steve Arcone (CRREL)

1/2014-12/2014 NASA Terrestrial Hydrology Program (THP)

NASA Snow School

Co-PI, $65,340

PI: Mark Sereze (National Snow and Ice Data Center), Co-PIs: Michael Durand (Ohio State University), Kelly Elder (USFS), Jessica Lundquist (University of Washington), Noah Molotch (University of Colorado), Matthew Sturm (University of Alaska, Fairbanks)

01/2014-12/2014 International Association of Oil and Gas Producers (OGP)

Program Design and Management for Remote Sensing Tests with Oil in Ice

Co-PI, $1,409,449 total, BSU portion: $128,731

BSU PI: John Bradford

Collaborators: Oil Spill Recovery Institute, University of Alaska Fairbanks, Prince William Sound Science Center, U.S. Army Cold Regions Research and Engineering Laboratory (CRREL), Polar Ocean Services, Woods Hole Oceanographic Institute, Norbit, Inland Gulf Marine, University of Washington

09/2012-08/2013 NSF Major Research Instrumentation

MRI: Acquisition of a GPU-accelerated High Performance Computing and Visualization Cluster

Co-PI, $555,384

PI: Inanc Senocak, Co-PIs: Peter Mullner, Jodi Mead, Tim Anderson (BSU)

09/2012-08/2013 NSF Major Research Instrumentation

MRI: Development of a laser-ultrasonic ice core tomography system
Co-PI, $336,122
PI: Kasper Van Wijk (BSU)

04/2012-04/2013 NSF Hydrologic Sciences

RAPID: An unusual opportunity to track snow ablation using stable isotope evolution of the 2011-2012 snowpack near Boise, Idaho
Co-PI, $19,912
PI: Sam Evans, Co-PIs: Alejandro Flores, Matt Kohn (BSU)

09/2012-08/2015 Idaho Department of Transportation

Development of an operational infrasonic avalanche detection system for Highway 21, Idaho
Principal Investigator, $100,000
Co-PIs: Scott Havens, Jeff Johnson (BSU)

01/2011-12/2013 NASA Experimental Program to Stimulate Competitive Research (EPSCoR)

Remote sensing of the Cryosphere: calibration and validation
Science Principal Investigator, $1,124,552 ($374,932 university match)
Co-PI: John Bradford, Kasper Van Wijk, Jim McNamara, and
Alejandro Flores, BSU Geosciences (BSU)

09/2010-08/2013 NASA New Investigator Program

Improvement of remotely sensed snow water equivalent estimates from microwave radar
Principal Investigator, $329,953 (1st in Idaho history)

04/2010-03/2013 NSF Hydrological Sciences Program

Quantifying lateral flow of water in alpine snowpacks using high-resolution geophysical techniques
Principal Investigator, $228,963
Co-PI: John Bradford and Jim McNamara, BSU

09/2010-12/2013 NASA Earth and Space Science Fellowship

Characterizing snow microstructure with X-ray tomography and a snow micropenetrometer: understanding important effects of microstructure on remote sensing
Principal Investigator, $90,000
Student: Scott Havens

01/2010-12/2012 DF Dickins Associates and consortium of oil companies

Detecting oil on and under sea ice using ground-penetrating radar: proposal for developing a new airborne system
Principal Investigator, $328,747
Co-PI: John Bradford, BSU

09/2010-08/2011 Idaho Department of Transportation Research Program

Development of an avalanche forecasting model for Highway 12 and 21

Principal Investigator, $10,000

11/2011-12/2014 NASA Instrument Incubator Program (IIP) - Phase 1

An 8-40 GHz Wideband Instrument for Snow Measurements (WISM)
Principal Investigator for BSU subcontract, BSU portion $94,947
Collaborative proposal with Harris Corporation, NASA Goddard,
University of Washington, Nuvotronics LLC

08/2010-08/2012 NASA Terrestrial Hydrology Program

Ground-based radar calibration / validation measurements for NASA CLPX-II 2010, Grand Mesa, Colorado
Principal Investigator, $35,468

08/2006-07/2009 NASA Energy and Water-cycle Sponsored (NEWS) Research and Terrestrial Hydrolology Program (THP)

Investigation of Spatial and Temporal Variation in Snowpack Properties Using Ground-Based High Resolution Microwave Radar Combined with Detailed Snowpack Characterization
Principal Investigator, $462,132 ($114,232 moved to BSU when hired)
Co-I: Nick Rutter (Northumbria University, U.K), Rick Forster (University of Utah)

09/2005-08/2010 NSF Arctic Sciences Program

Collaborative Research: Polarimetric Characteristics of Radio-wave Scattering from Water Pathways within Glaciers: Laboratory Experiments and Computer Simulations
Principal Investigator, $69,785
Co-PI: Kenichi Matsuoka (University of Washington)

Advising

Primary Advising

Name	**Degree**	**Advisor Role**	**Program/University**	**Graduation**
Achim Heilig	Post-Doc	Primary	Geophysics/BSU	01/11-12/12
Scott Havens	Ph.D.	Primary	Geophysics/BSU	Fall 2014
Andrew Hedrick	Ph.D.	Primary	Geophysics/BSU	Fall 2018
Joel Gongora	Ph.D.	Primary	Geophysics/BSU	exp. Spring 2021
Tate Meehan	Ph.D.	Primary	Geophysics/BSU	exp. Spring 2021
Naheem Adebisi	Ph.D.	Primary	Geophysics/BSU	exp. Spring 2024
Zach Keiskinen	Ph.D.	Co-Primary	Geophysics/BSU	exp. Spring 2024
Evi Ofekeze	Ph.D.	Co-Primary	Geophysics/BSU	exp. Spring 2023
Ibrahim Alabi	Ph.D.	Co-Primary	Geophysics/BSU	exp. Spring 2023
Chago Rodriguez	Ph.D.	Primary	Geophysics/BSU	exp. Spring 2022
Mitch Creelman	M.S.	Primary	Geosciences/BSU	exp. Fall 2021
Megan Mason	M.S.	Primary	Geophysics/BSU	Summer 2020
Peter Youngblood	M.S.	Primary	Hydrology/BSU	Summer 2020
Tate Meehan	M.S.	Co-primary	Geophysics/BSU	Spring 2018
Andrew Hedrick	M.S.	Primary	Geophysics/BSU	Fall 2013
Jesse Dean	M.S.	Primary	Geophysics/BSU	Fall 2016
Mark Robertson	M.S.	Primary	Geophysics/BSU	Fall 2016
Gabriel Trisca	M.S.	Primary	Computer Science/BSU	Spring 2014
Tom Pe	Undergrad	Primary	Applied Physics/BSU	Spring 2019
Peter Olsoy	Undergrad	Primary	Biology/BSU	Spring 2011
Jennifer Morris	Undergrad	Primary	Geosciences/BSU	Spring 2011
Jason Jennings	Undergrad	Primary	Geophysics/BSU	Spring 2012
Andrew Peterson	Undergrad	Primary	Geophysics/BSU	Spring 2012
Jake Robertson	Undergrad	Primary	Geosciences/BSU	Spring 2013
Rachel Parsons	Undergrad	Primary	Geosciences/BSU	Spring 2013
Mark Robertson	Undergrad	Primary	Geophysics/BSU	Fall 2012
Andrew Karlson	Undergrad	Primary	Geosciences/BSU	Spring 2016
Robert Florence	Undergrad	Primary	Geosciences/BSU	Spring 2015
Kris Burch	Undergrad	Primary	ECE/BSU	Spring 2012
Lloyd Lowe II	Undergrad	Primary	ECE/BSU	Spring 2012
Kolton Drake	Undergrad	Primary	ECE/BSU	Spring 2012

Committee Member / Secondary Advising

Name	**Degree**	**Advisor Role**	**Program/University**	**Graduation**
Arash Modaresi Rad	Ph. D.	Committee	PhD Computing	Spring 2023
Jukes Lui	M.S.	Committee	Geosciences/BSU	Summer 2020
Colton Elkin	M.S.	Committee	Geosciences/BSU	Spring 2021
Rainey Aberle	M.S.	Committee	Geosciences/BSU	Spring 2021
Kate Bollen	M.S.	Committee	Geosciences/BSU	Spring 2021
Josh Morrel	M.S.	Committee	Geosciences/BSU	Spring 2021
Allison Vincent	M.S.	Committee	Geosciences/BSU	Summer 2020
Jewell Lund	M.S.	Committee	Geography/Univ Utah	Spring 2021
Will Rudisill	Ph.D.	Committee	Geosciences/BSU	Spring 2023
Ahmad Hojatimalek	Ph.D.	Committee	Geosciences/BSU	Spring 2023
Maggi Kraft	Ph.D.	Committee	Geosciences/BSU	Spring 2022
Buchanan Kerswell	Ph.D.	Committee	Geosciences/BSU	Spring 2021
Miguel Aguayo	Ph.D.	Committee	Geosciences/BSU	Summer 2018
Katelyn Watson	Ph.D.	Committee	Geosciences/BSU	exp. Summer 2020
Jake Anderson	Ph.D.	Committee	Geophysics/BSU	Spring 2019
Kara Ferguson	M.S.	Committee	Geosciences/BSU	Spring 2018
Zakk Hess	M.S.	Committee	Geosciences/BSU	exp. Fall 2018
Zach Uhlmann	M.S.	Committee	Geosciences/BSU	Summer 2018
Megan Gallagher	M.S.	Committee	Geophysics/BSU	Summer 2018
William Rudisill	M.S.	Committee	Geosciences/BSU	Spring 2018
Lacie Rasley	M.S.	Committee	Geosciences/BSU	exp Spring 2018
Eric Lutz	Ph.D.	Committee	Geosci./Montana State Univ.	Spring 2009
James McCreight	Ph.D.	Committee	Civil Eng./Univ. Colorado	Fall 2011
Dylan Mikesell	Ph.D.	Committee	Geophysics/BSU	Fall 2011
Joel Brown	Ph.D.	Committee	Geophysics/BSU	Fall 2011
Eli Deeb	Ph.D.	Committee	Geography/Univ. Utah	Fall 2012
Thomas Blum	Ph.D.	Committee	Geophysics/BSU	Spring 2013
Patrick Kormos	Ph.D.	Committee	Geosciences/BSU	Fall 2013
Wade Tinkham	Ph.D.	Committee	Forest Ecol., Biogeosci./U of I	Spring 2013
Nicolas Kinar	Ph.D.	External Examiner	Hydrology, U of Saskatchewan	Fall 2013
Esther Babcock	Ph.D.	Committee	Geophysics/BSU	Spring 2015
Brian Terbush	M.S.	Committee	Geosciences/BSU	Spring 2015
Ravi Preeshageetha	M.S.	Committee	Computer Science/ BSU	Spring 2015
Josh Nichols	M.S.	Committee	Geophysics/BSU	Spring 2010
Emily Park	M.S.	Committee	Geophysics/BSU	Spring 2011
Alden Shallcross	M.S.	Committee	Geosciences/BSU	Fall 2011
Brian Anderson	M.S.	Committee	Geosciences/BSU	Spring 2011
David Eiriksson	M.S.	Committee	Geosciences/BSU	Spring 2012
Reggie Walters	M.S.	Committee	Geosciences/BSU	Fall 2013
Erik Boe	M.S.	Committee	Geosciences/BSU	Fall 2013

Teaching - BSU Courses Taught

Course Number/Name	**Semester**	**Credits**	**# grad/ugrad**
GEOPH 567: Snow Science Field Methods	Spring 2022	2	10/0
GEOPH 422/522: Data Analysis and Geostatistics	Fall 2021	3	20/3
GEOPH 566: Snow and Ice Physics	Spring 2021	3	6/0
GEOPH 422/522: Data Analysis and Geostatistics	Fall 2020	3	24/5
GEOPH 597: CUAHSI Virtual University	Fall 2020	3	4/0
GEOPH 567: Snow Science Field Methods	Spring 2020	2	1/0
GEOPH 597: CUAHSI Virtual University	Fall 2019	3	10/0
GEOPH 597: Select Topics in Snow Science	Spring 2019	1	5/0
GEOPH 422/522: Data Analysis and Geostatistics	Fall 2018	3	13/0
GEOPH 480: Research in Geophysics	Spring 2018	3	0/1
GEOPH 466/566: Snow and Ice Physics	Spring 2018	3	6/7
GEOPH 422/522: Data Analysis and Geostatistics	Fall 2017	3	10/2
GEOPH 422/522: Data Analysis and Geostatistics	Fall 2016	3	11/3
GEOS 486: Geosciences Capstone	Spring 2016	4	0/1
GEOPH 466/566: Snow and Ice Physics	Spring 2016	3	4/0
GEOPH 567: Snow Science Field Methods	Spring 2016	2	1/0
GEOPH 422/522: Data Analysis and Geostatistics	Fall 2014	3	12/1
GEOPH 502: Processes and Properties of the Earth II	Spring 2014	4	4/0
GEOPH 467/567: Snow Science Field Methods	Spring 2014	2	1/1
GEOPH 480: Research in Geophysics	Spring 2014	3	0/1
GEOPH 422/522: Data Analysis and Geostatistics	Fall 2013	3	16/0
GEOPH 502: Properties and Processes of the Earth II	Spring 2013	4	5/0
GEOPH 467/567: Snow Science Field Methods	Spring 2013	2	2/1
GEOPH 466/566: Snow and Ice Physics	Spring 2012	3	9/4
GEOPH 480: Research in Geophysics	Spring 2012	2	0/2
GEOPH 422/522: Data Analysis and Geostatistics	Fall 2011	3	9/2
GEOPH 497/597: Snow Radar Remote Sensing	Fall 2011	1	7/2
GEOPH 480: Research in Geophysics	Fall 2011	2	0/2
GEOPH 497/597: Snow Science Field Methods	Spring 2011	2	5/7
GEOPH 497/597: Data Analysis and Geostatistics	Fall 2010	3	11/2
GEOPH 497/597: Snow and Ice Physics	Spring 2010	3	7/2
GEOPH 497/597, CE 497/597: Data Analysis / Geostat.	Fall 2009	3	18/1
GEOPH 305: Applied Geophysics, w/ John Bradford	Spring 2009	3	0/5
TOTAL		**90**	**231/55**

Teaching - Other Courses

01/2003-06/2003 T.A., *The Weather*, Program on Ocean and Atmospheric Sciences, Univ. CO

01/2000-06/2000 T.A., *Engineering Geology*, Dept. of Civil Engineering, Univ. CO

12/2002-12/2003 Instructor, *Avalanche Level I*, Alpine World Ascents

08/2001-06/2002 N.S.F. K-12 Outreach Fellow

Provided lectures/hands-on demonstrations to local 7th and 8th grade classes on glaciology and avalanche related topics; communicated via website with students during field season in W. Antarctica. Supervised high-school student on avalanche-related senior research project.

09/1999-03/2000 Volunteer High School Teacher, Old Crow, Yukon Territory

Developed and taught individualized curriculum for at-risk high school youth in this small isolated Arctic village of Gwitch'in First Nation people. Supervised construction on Youth Center, obtained funding for and organized week-long hunting and cultural activities field trip, created afterschool and evening basketball program. (see also Honors / Awards)

University Service

8/2018-present	Steering committee for the Ph.D in Computing
6/2019-present	Tenure and Promotion Committee, Geoscience Dept
8/2019-8/2021	Tenure and Promotion Committee, College of Arts and Sciences
1/2019-6/2019	Graduate Program Committee, Geophysics Rep.
08/2012-12/2015	Graduate Program Committee, Geophysics Rep.
2011-2019	Four hiring committees, Geophysics Faculty
09/2008-09/2011	Undergraduate Cirriculum Committee, Geophysics Rep.
2012	Hiring committee, COAS Scientific Instrumentation Shop, Geosciences rep.
Fall 2010	Organized installation of INL software in Mini-Cave 3-D Visualization Laboratory
01/09-present	Established geodetic/survey grade GPS instrument pool for department
2010	Co-PI, BSU designated a NVIDIA CUDA HPC research center (one of 20 worldwide)
2011-present	Organize avalanche training and maintain pool of avalanche safety equipment for BSU

External Service

2021-present	NASA SnowEx Mission, Leadership team member
2021-present	Chair of AGU Cryosphere DEI Committee
2019-2021	NASA SnowEx Mission, Lead Project Scientist
2017	NASA SnowEx Mission, Leadership Team and co-lead of field campaign
2016	NASA Earth Venture Mission Panel Member (proposals for future Earth observation satellites)
01/2015-present	Education board member, NASA international Snow Working Group for

	Remote Sensing (iSWGR)
02/2015, 01/2016	Instructor, NASA iSWGR field school
12/2008-present	Session Chair, Fall AGU Meeting (16 total sessions 2008-2012)
Jan 6-8, 2012	Co-organizer, *SEG Cryosphere Geophysics Workshop: Understanding a Changing Climate with Subsurface Imaging*, Boise, ID
12/2008-01/2012	Cryosphere Section Representative, Fall AGU meeting
	2 trips per year to Washington, D.C. to organize Cryosphere program (> 500 submissions annually) for largest geophysics meeting in U.S. (>15,000 submissions)
2011	NASA Senior Review Panel Member (review of all 13 active Earth Science NASA Missions)
2010-2011	Lead Guest Editor: CRST special issue for the *2010 International Snow Science Workshop* (premier snow science meeting worldwide)
12/2010	Invited Keynote Speaker at the PRYN/APECS Career Development Workshop
	Involving the next generation in Cryospheric Sciences: Getting advising and mentoring experience during graduate school through participation in undergraduate and high school research programs
03/2008,04/2009	Co-organized and taught Snow Characterization Workshop with Dr. Martin Schneebeli, Swiss Federal Institute for Snow and Avalanche Research (SLF)
	3 day classroom and field-based workshop held in Silverton, Colorado in 2008 at the Center for Snow and Avalanche Studies, 10 invited graduate students and scientists from different U.S. universities. Invited to repeat workshop in 2009 in Manali, India, by the Indian Department of Defense Snow and Avalanche Study Establishment (SASE), more than 30 Indian Ph.D. scientists attended.
01/2005-08/2011	Chair of Research Committee, American Avalanche Association
	Administered four grants per year, only funding source specifically for snow avalanche research in U.S.
01/2004-present	Journal Reviewer: *Cold Regions Science and Technology; Hydrologic Processes; Journal of Geophysical Research - Earth Surface; Journal of Geophysical Research - Atmospheres; Annals of Glaciology; Water Resources Research; Earth Surface Processes and Landforms; Journal of Glaciology; Transactions on Geoscience and Remote Sensing; Remote Sensing of Environment; Arctic, Antarctic, and Alpine Research; Geoscience and Remote Sensing Letters; Near Surface Geophysics; Northwest Science; Journal of Applied Geophysics*
04/2008-present	Proposal Reviewer: *NASA Terrestrial Hydrology Program (panel member); NSF Experimental Program to Stimulate Competitive Research (EPSCoR) program; NSF Arctic Sciences; NASA Cryospheric Sciences IceBridge (panel member); NSF Geophysics; NSF Hydrologic Sciences; American Avalanche Association*
2010	Organized and delivered *Snow Day* at the Discovery Center of Idaho
2011-present	Train volunteers and develop curriculum for Bogus Basin *Snow School*
06/2007-09/2009	Research Committee, Eastern Snow Conference
10/2009-09/2012	Research committee, American Institute for Avalanche Research and Education
09/2006	Session Chair, Fracture Mechanics, International Snow Science Workshop

also reviewed papers and helped plan this international conference; more than 800 total attendees

2001-2005 Senior Research Project Mentor for 6 Boulder Valley HS students

Awards: 2002: Earth Sciences 1st Place, State and Nat. Finalist, Best in Show (Adam Maerz); 2001: Earth Sciences 3rd Place (John Powers)

Conference Proceedings Papers

2018 Johnson, J. B., **H.-P. Marshall**, Loo, S. M., Nalli, B., Saurer, M., Havens, S., Colton, J., and Anderson, J. F. (2018). Detection and tracking of snow avalanches in little cottonwood canyon, utah using multiple small-aperture infrasound arrays. *Proceedings of the 2018 International Snow Science Workshop*

Rodriguez, S. and **H.-P. Marshall** (2018). Application of a k-band microwave sensor in the detection of water melt-freeze states within a snowpack. *Proceedings of the 2018 International Snow Science Workshop*

2016 **H.-P. Marshall** (2016). Evaluation of avatech sp2 for snow research and snow science education. *Proceedings of the 2016 International Snow Science Workshop*

H.-P. Marshall and Robertson, M. (2016). Continuous real-time snow properties with ground penetrating radar. *Proceedings of the 2016 International Snow Science Workshop*

2014 **Marshall, H.P.** and CryoGARS (2014). Water in snow likes to go with the flow: Dynamics of liquid water in snow and its impact on stability. *Proceedings of the 2014 International Snow Science Workshop*

Dean, J., **H.-P. Marshall**, Lutz, E., Christian, J., Markle, B., and Whitmore, S. (2014). Using nir reflectance for characterizing stratigraphy and liquid water in snow. *Proceedings of the 2014 International Snow Science Workshop*

Elder, K., **H.-P. Marshall**, Elder, L., Starr, B., Karlson, A., and Robertson, J. (2014). Design and installation of a tipping bucket snow lysimeter. *Proceedings of the 2014 International Snow Science Workshop*

Havens, S., **H.-P. Marshall**, Johnson, J., and Nicholson, B. (2014a). Calculating the velocity of a fast moving snow avalanche using an infrasound array. *Proceedings of the 2014 International Snow Science Workshop*

Hedrick, A. and **H.-P. Marshall** (2014). Automated snow depth measurements in avalanche terrain using time-lapse photography. *Proceedings of the 2014 International Snow Science Workshop*

Lutz, E. and **H.-P. Marshall** (2014). Validation study of avatech's rapid snow penetrometer, sp1. *Proceedings of the 2014 International Snow Science Workshop*

Robertson, M., **H.-P. Marshall**, Roberts, D., and Tidd, W. (2014). Measuring snow water equivalent in wet snow with ultrawideband gpr. *Proceedings of the 2014 International Snow Science Workshop*

Rodriguez, S., **H.-P. Marshall**, and Rodriguez, P. (2014). Applications of low cost and low power fmcw radar in the snow characterization of dry snow. *Proceedings of the 2014 International Snow Science Workshop*

2012 **H.-P. Marshall**, , Gleason, A., Heilig, A., Koh, G., and Sturm, M. (2012). Spatial variability of snow stratigraphy from microwave radar. *Proceedings of the 2012 International Snow Science Workshop*

Havens, S., **H.-P. Marshall**, Covington, C., Nicholson, B., and Conway, H. (2012). Real time slope stablity modeling of direct avalanches. *Proceedings of the 2012 International Snow Science Workshop*

Hedrick, A., **H.-P. Marshall**, Heilig, A., and Winstral, A. (2012). Testing and implementing a wind redistribution model in mountainous terrain using a high-resolution LiDAR-derived DEM. *Proceedings of the 2012 International Snow Science Workshop*

Heilig, A., **H.-P. Marshall**, and Evans, S. (2012). Temporal and spatial changes of a seasonal snowpack on a small sheltered slope. *Proceedings of the 2012 International Snow Science Workshop*

Elder, K., Angutikjuak, I., Baker, J., Belford, M., Bennett, T., Birkeland, K., Bowker, D., Chabot, D., Cheuvront, A., Dixon, M., Elder, D., Elder, L., Gearheard, S., Giedt, G., Grant, K., Green, S., Greene, E., Houfek, N., Huntington, C., Huntington, H., Huntington, T., Janigian, D., Johnson, C., Liston, G., Maris, R., Marsh, A., **H.-P. Marshall**, Meiners, A., Meiners, A., Meiners, T., Palluq, L., Pope, J., Qillaq, E., Sanguya, J., Sehnert, S., Simenhois, R., Starr, B., and Tyler, R. (2012). Meteorological tower design for severe weather and remote locations. *Proceedings of the 2012 International Snow Science Workshop*

2010 Havens, S., **H.-P. Marshall**, Steiner, N., and Tedesco, M. (2010). Snow micro penetrometer and near infrared photography for grain type classification. *Proceedings of the 2010 International Snow Science Workshop*

2009 Heilig, A., **H.-P. Marshall**, Eisen, O., and Schneebeli, M. (2009). Non-destructive quantification of snowpack properties. *Proceedings of the 2009 International Snow Science Workshop*

2008 **H.-P. Marshall**, Birkeland, K., Elder, K., and Meiners, T. (2008a). Helicopter-based microwave radar measurements in alpine terrain. *Proceedings of the 2008 International Snow Science Workshop*

H.-P. Marshall, Landry, C., Hale, S., Roberts, J., and Conway, H. (2008d). Snow slope stability modeling of direct-action avalanches in a continental climate: Red mountain pass, co. *Proceedings of the 2008 International Snow Science Workshop*

H.-P. Marshall, Koh, G., and Sturm, M. (2008c). Ultra-broadband portable microwave fmcw radars for measuring snow depth, snow water equivalent, and stratigraphy: practical considerations. *International Union of Radio Science (URSI) General Assembly.* (Invited)

Pielmeier, C. and **H.-P. Marshall** (2008). Estimating rutschblock stability from snowmicropen measurements. *Proceedings of the 2008 International Snow Science Workshop*

Gleason, A., **H.-P. Marshall**, and McCreight, J. (2008). Into the belly of the beast (snow depth across starting zones measured with frequency modulated continuous wave radar). *Proceedings of the 2008 International Snow Science Workshop*

Birkeland, E. L. K., **H.-P. Marshall**, and Hansen, K. (2008). Quantifying changes in weak layer microstructure associated with loading events. *Proceedings of the 2008 International Snow Science Workshop*

2007 **H.-P. Marshall**, Rutter, N., Koh, G., O'Neel, S., and McCreight, J. (2007). Ground-based fmcw radar measurements of dry snowpacks during the 2006-07 nasa clpx field experiment. *Eastern Snow Conference*

Demuth, M., **H.-P. Marshall**, Morris, E., Burgess, D., and Bell, C. (2007). High-resolution near-surface snow stratigraphy inferred from ground-based broadband microwave radar measurements: Devon ice cap, nunavut, canada 2005-06, cryosat/asiras validation campaign. *Eastern Snow Conference*

2006 Demuth, M., Morris, E., **H.-P. Marshall**, Fisher, D., Sekerka, J., Koerner, R., and Gray, L. (2006). Characterizing percolation and wet-snow facies variability on devon ice cap, nunavut, canada. *Proceedings of the 2006 International Snow Science Workshop*

H.-P. Marshall, Koh, G., Sturm, M., Johnson, J., Demuth, M., Landry, C., Deems, J., and Gleason, A. (2006a). Spatial variability of the snowpack: experiences with measurements at a wide range of length scales with several different high precision instruments. *Proceedings of the 2006 International Snow Science Workshop*

Gleason, A. and **H.-P. Marshall** (2006). Deformation analysis of the snow micropen using particle image velocimetry. *Proceedings of the 2006 International Snow Science Workshop*

Pielmeier, C., Schweizer, J., and **H.-P. Marshall** (2006). Improvements in the application of the snowmicropen to derive stability information for avalanche forecasting. *Proceedings of the 2006 International Snow Science Workshop*

2005 Koh, G. and **H.-P. Marshall** (2005). Radar signature of recently disturbed soil for mine detection. *SPIE - International Society for Optical Engineering, Session on Detection andf Remediation Technologies for Mines and Mine like Targets*

2004 Hayes, P., Wilbour, C., Gibson, R., **H.-P. Marshall**, and Conway, H. (2004). A simple model of snow slope stability during storms. *Proceedings of the 2004 International Snow Science Workshop*

H.-P. Marshall, Schneebeli, M., Koh, G., Matzl, M., and Pielmeier, C. (2004). Measurements of snow stratigraphy with fmcw radar: comparison with other snow science instruments. *Proceedings of the 2004 International Snow Science Workshop*

Gleason, J. and **H.-P. Marshall** (2004). Measurements of snow strain combined with finite element simulations. *Proceedings of the 2004 International Snow Science Workshop*

2002 **H.-P. Marshall** (2002a). Modeling the interface between two layers of snow. *Proceedings of the 2002 International Snow Science Workshop*

1998 **H.-P. Marshall**, Conway, H., and Rasmussen, L. (1998). Snow densification during rain. *Proceedings of the 1998 International Snow Science Workshop*

AGU Talks

2019 **H.-P. Marshall**, Deeb, E., Lavalle, M., Elder, K., Brucker, L., Hiemstra, C., Bormann, K., Painter, T., Forster, R., and Jones, C. (2019). Airborne snow accumulation estimates from L-band InSAR during the NASA SnowEx 2017 campaign and validation with airborne lidar and in-situ observations. *Fall AGU, San Francisco, CA, Abstract C42B-08*

Johnson, J., Anderson, J., Angelis, S. D., Escobar-Wolf, R., Lyons, J., **H.-P. Marshall**, and Pineda, A. (2019). On the capabilities of networked infrasound arrays for investigating rapid gravity-driven mass movements: lahars and snow avalanches. *Fall AGU, San Francisco, CA, Abstract V44B-02*

Lewis, G., Osterberg, E. C., Hawley, R., **H.-P. Marshall**, Meehan, T., Graeter, K., McCarthy, F., Overly, T., Thundercloud, Z., Ferris, D., Birkel, S., Dibb, J., Koffman, B., and Tedesco, M. (2019b). Climatic controls on albedo across the West Greenland ice sheet percolation zone. *Fall AGU, San Francisco, CA, Abstract C33A-05*

Osterberg, E. C., Lewis, G., Hawley, R., **H.-P. Marshall**, Ferris, D., Thundercloud, Z., Graeter, K. A., DeAngelo, J., Meehan, T., Overly, T. B., Birkel, S., and McCarthy, F. (2019). Atmospheric blocking is a critical but poorly understood driver of summertime Greenland surface mass balance. *Fall AGU, San Francisco, CA, Abstract C34A-03*

Deeb, E., **Marshall, H.P.**, Courville, Z., Lever, J., Forster, R., and Shoop, S. (2019). Remote snow strength detection using multi-frequency / multi-polarization radar. *Fall AGU, San Francisco, CA, Abstract C41B-04*

Webb, R., Raleigh, M., McGrath, D., Molotch, N., Elder, K., Hiemstra, C., Brucker, L., and **H.-P. Marshall** (2019). Within-stand boundary effects of snow water equivalent distribution in forested areas. *Fall AGU, San Francisco, CA, Abstract C41B-08*

2018 Marshall, H.-P., Deems, J., Deeb, E., Liston, G., Hall, D., Gatebe, C., Derksen, C., Tsang, L., Kim, E., Painter, T., Sandells, M., Houser, P., Molotch, N., and Reinking, A. (2018). Snowex 2019 - a time series airborne and field experiment in multiple snow climates. *Fall AGU, Washington, D.C., Abstract C12A-01*

Deeb, E., Marshall, H., Forster, R., Jones, C., and Lavalle, M. (2018). Polarimetric and interferometric analysis of l-band airborne data over western united states during the nasa snowex 2017 field campaign. *Fall AGU, Washington, D.C., Abstract C12A-07*

Gongora, J., Marshall, H., Glenn, N., Flores, A., and Mead, J. (2018). Deep and shallow learning in deep and shallow snow. *Fall AGU, Washington, D.C., Abstract H34B-08B*

Lewis, G., Osterberg, E. C., Hawley, R., Marshall, H., Birkel, S. D., Dibb, J., Koffman, B., Ferris, D., and Tedesco, M. (2018). Effects of mineral dust and black carbon on albedo in the western greenland ice sheet percolation zone. *Fall AGU, Washington, D.C., Abstract C31A-05*

Lievens, H., Demuzere, M., DeLannoy, G., and Marshall, H. (2018). Mapping snow mass in the northern hemisphere mountains with sentinel-1. *Fall AGU, Washington, D.C., Abstract C11B-07*

Lundquist, J., Deems, J., Durand, M., Langlois, A., Marshall, H., Raleigh, M., Sandells, M., Sturm, M., and Vuyovich, C. (2018). A strategy for global snow information for nature and society (invited). *Fall AGU, Washington, D.C., Abstract H22B-04*

2017 Osterberg, E. C., Graeter, K., Hawley, R. L., Marshall, H.-P., Ferris, D. G., Lewis, G., Birkel, S. D., Meehan, T., and McCarthy, F. (2017). Recent rise in west greenland surface melt and firn density driven by north atlantic ssts and blocking events. *Fall AGU, New Orleans, LA, Abstract C24C-08*

McGrath, D., Webb, R., **H.-P. Marshall**, Hale, K., and Molotch, N. (2017). Spatially extensive ground-penetrating radar snow depth observations during nasa's 2017 snowex campaign. *Fall AGU, New Orleans, LA, Abstract C11F-01*

McNamara, J., Benner, S., Chandler, D., Flores, A., **H.-P. Marshall**, Seyfried, M., Poulos, M., and Pierce, J. (2017). Form and function relationships revealed by long-term research in a semiarid mountain catchment. *Fall AGU, New Orleans, LA, Abstract H43O-04*

2016 McGrath, D., Sass, L., McNeil, C., Babcock, E., Candela, S., O'Neel, S., Arendt, A., and **H.-P. Marshall** (2016). Inter-annual variability in snow accumulation using ground-penetrating radar on Wolverine and Gulkana glaciers, Alaska: Implications for glacier mass balance modeling. *Fall AGU, San Francisco, CA, Abstract C14C-01*

O'Neel, S., Sass, L., Koch, J., Littell, J., Klein, E., **H.-P. Marshall**, LeWinter, A., Larsen, C., Hill, D., Hood, E., Sawicki, S., and Carter, A. (2016). From icefield to ocean: investigating biophysical linkages at Wolverine Glacier, Alaska. *Fall AGU, San Francisco, CA, Abstract H11D-04*

2015 McNamara, J. P., **Marshall, H.P.**, Kohn, M., Evans, S., and Flores, A. (2015). Lateral flow in snow as a runoff generation mechanism. *Fall Meeting, AGU, San Francisco, CA, Abstract C44B-01*

Lewis, G., Hawley, R., Osterberg, E., and **Marshall, H.P.** (2015). Icebridge radar as a tool for understanding accumulation variability throughout the western greenland ice sheet. *Fall Meeting, AGU, San Francisco, CA, Abstract C51E-04*

Deeb, E., **Marshall, H.P.**, Forster, R. R., Arcone, S., Siqueira, P., and Morriss, B. (2015). The present and future of using insar to estimate swe. *Fall Meeting, AGU, San Francisco, CA, Abstract C31B-04*

2014 **Marshall, H.P.**, Heilig, A., Evans, S., Robertson, M., Hetrick, H., Eiriksson, D., Dean, J., Karlson, A., Hedrick, A., Bradford, J. H., McNamara, J. P., Flores, A., Kohn, M., and Rodriguez, C. (2014). Liquid water dynamics in unsaturated snow: the role of lateral flow. *Fall Meeting, AGU, San Francisco, CA, Abstract C22B-07*

Hedrick, A., **Marshall, H.P.**, Winstral, A., Elder, K., Yueh, S., and Cline, D. (2014b). Using regional-scale lidar surveys to validate operational snow models. *Fall Meeting, AGU, San Francisco, CA, Abstract C32A-06*

Deeb, E., **Marshall, H.P.**, Lamie, N., and Arcone, S. (2014). Quantifying the accuracy of snow water equivalent estimates using broadband radar signal phase. *Fall Meeting, AGU, San Francisco, CA, Abstract C31E-05*

Havens, S., **Marshall, H.P.**, Trisca, G., Johnson, J. B., and Nicholson, B. (2014d). Modeling and monitoring avalanches caused by rain-on-snow events. *Fall Meeting, AGU, San Francisco, CA, Abstract C31E-03*

McNamara, J. P., Aishlin, P., Flores, A., Benner, S. G., **Marshall, H.P.**, and Pierce, J. (2014). Integrating local research watersheds into hydrologic education: Lessons from the dry creek experimental watershed. *Fall Meeting, AGU, San Francisco, CA, Abstract ED54A-3508*

2013 Robertson, M., Koenig, L., Trisca, G., and **Marshall, H.P.** (2013). Accumulation mapping at summit, greenland using an autonomous rover (invited). *Fall Meeting, AGU, San Francisco, CA, Abstract C24B-07*

2012 Gusmeroli, A., Wolken, G., Arendt, A. A., Campbell, S., O'Neel, S., and **Marshall, H.P.** (2012). Ground-penetrating radar observations of winter snow accumulation on alaska glaciers. (invited). *Fall AGU, San Francisco, CA, Abstract C52A-03*

2011 Alexander, P., Tedesco, M., Steiner, N., **H.-P. Marshall**, Luthcke, S. B., and Fettweis, X. (2011). Identification of accumulation, density and grain size bias in the regional climate model mar over the greenland ice sheet using in-situ and remotely sensed data. *Fall Meeting, AGU, San Francisco, CA, Abstract C22B-06*

2010 **H.-P. Marshall**, Marks, D. G., and Hudak, A. H. W. R. S. A. T. (2010). Estimating swe distribution with a combination of ground-based radar measurements, modeling and remote sensing. *Fall AGU, San Francisco, CA, Abstract C11C-01.* (Invited)

Yueh, S. H., Rott, H., Nagler, T. F., Cline, D. W., Duguay, C. R., Essery, R., Etchevers, P., Hajnsek, I., Kern, M., Macelloni, G., Malnes, E., Pulliainen, J. T., Tsang, L., Xu, X., **H.-P. Marshall**, and Elder, K. (2010). Microwave radar retrieval of snow water equivalent. *Fall Meeting, AGU, San Francisco, CA, Abstract C54B-03*

Bradford, J. H. and **H.-P. Marshall** (2010). Estimating complex dielectric permittivity of soils from spectral ratio analysis of swept frequency (FMCW) ground-penetrating radar data. *Fall Meeting, AGU, San Francisco, CA, Abstract H21K-06.* (Invited)

Johnson, J. B., Marcillo, O. E., Arechiga, R. O., Johnson, R., Edens, H. E., **H.-P. Marshall**, Havens, S., and Waite, G. P. (2010). Investigation of the infrasound produced by geophysical events such as volcanoes, thunder, and avalanches: the case for local infrasound monitoring. *Fall Meeting, AGU, San Francisco, CA, Abstract S14A-01.* (Invited)

2009 **H.-P. Marshall**, Demuth, M. N., Gray, L., Burgess, D. O., and Morris, E. M. (2009a). High-resolution annual accumulation rate measurements using FMCW radar, Devon Ice Cap, Nunavut, Canada. *EOS Trans. AGU Fall meeting suppl., Abstract C34B-01*, 90. (Invited)

McCreight, J. L., Rajagopalan, B., Slater, A. G., and **H.-P. Marshall** (2009). A comparison of methods for estimation and prediction of snow depth and its spatial non-stationarity at the first-order basin scale. *EOS Trans. AGU Fall meeting suppl., Abstract C21E-06*, 90. (Invited)

Lutz, E., Birkeland, K., **H.-P. Marshall**, and Hansen, K. (2009a). Spatial and temporal analysis of snowpack strength and stability and associations with micro-climate on an inclined forest opening. *EOS Trans. AGU Fall meeting suppl., Abstract C21E-08*, 90. (Invited)

2008 **H.-P. Marshall**, Rutter, N., Tape, K., Sturm, M., and Essery, R. (2008e). High resolution ground-based snow measurements during the NASA CLPX-II campaign, north slope, alaska. *EOS Trans. AGU Fall meeting suppl., Abstract C12B-05*, 89(53)

Rutter, N., **H.-P. Marshall**, Tape, K., and Essery, R. (2008). Observing microscale variations in snow stratigraphy using near infra-red photography. *EOS Trans. AGU Fall meeting suppl., Abstract C34A-07*, 89(53). (Invited)

2007 **H.-P.Marshall**, Koh, G., Sturm, M., and Rutter, N. (2007c). Estimating snow depth, snow water equivalent, and stratigraphy at high resolution using microwave radar. *EOS Trans. AGU Fall meeting suppl., Abstract H23H-04*, 88(52). (Invited)

Matsuoka, K. and **H.-P. Marshall** (2007). Laboratory and numerical experiments of radar back scattering: towards interpreting the shape of subsurface targets in cryospheric applications. *EOS Trans. AGU Fall meeting suppl., Abstract NS14A-04*, 88(52). (Invited)

2006 **H.-P. Marshall**, Sturm, M., Holmgren, J., and Koh, G. (2006b). Ground-based fmcw radar measurements for calibration/validation during amsr-ice06: Barrow, ak, usa. *EOS Trans. AGU Fall meeting suppl., Abstract C13A-07*, 87(52)

2005 McNamara, D., O'Neel, S., **H.-P. Marshall**, and Pfeffer, T. (2005). Iceberg calving at rapidly retreating tidewater glacier: Columbia glacier, ak, usa. *EOS Trans. AGU Fall meeting suppl., Abstract C51A-0261*, 86(52)

2003 **H.-P. Marshall**, Koh, G., and Forster, R. (2003). Ground-based fmcw radar measurements: a summary of the nasa clpx data. *EOS Trans. AGU Fall meeting suppl., Abstract C42B-07*, 84(46)

2001 Pfeffer, T., **H.-P. Marshall**, Harper, J., and Humphrey, N. (2001). The constitutive relationship of a glacial-scale block of temperate ice. *EOS Trans. AGU Fall meeting suppl., Abstract IP12B-04*, 82(47)

AGU Posters

2019 Kraft, M., McNamara, J., and **H.-P. Marshall** (2019). Measurement and modeling of snow surface net radiation under varying vegetation structures in a semi-arid mountain watershed. *Fall AGU, San Francisco, CA, Abstract C33B-1566*

Bonnell, R., McGrath, D., **H.-P. Marshall**, and Webb, R. (2019). Spatiotemporal variations in liquid water content in a continental seasonal snowpack measured by ground penetrating radar. *Fall AGU, San Francisco, CA, Abstract C33C-1600*

Marks, D., Havens, S., Hedrick, A., Trujillo, E., Pierson, F., Deems, J., Skiles, M., Bormann, K., Robertson, M., **H.-P. Marshall**, and Painter, T. (2019). The future of snow science: spatial data assimilation and modeling in mountain regions. *Fall AGU, San Francisco, CA, Abstract C33D-1608*

Hojatimalekshah, A., Uhlmann, Z., Glenn, N., Spaete, L., Hiemstra, C., Tennant, C., Gelvin, A., **H.-P. Marshall**, McNamara, J., Enterkine, J., and Graham, J. (2019). Examining multi-scale snow vegetation relationships using object-derived measures of canopy structure from terrestrial laser scanning, Grand Mesa, Colorado. *Fall AGU, San Francisco, CA, Abstract C33D-1613*

Vincent, A., Flores, A., Glenn, N., **H.-P. Marshall**, and Nash, C. (2019). Applying multisensor data fusion via the Spatial and Temporal Adaptive Reflectance Fusion Model (STARFM) for improved monitoring of seasonal snowpack. *Fall AGU, San Francisco, CA, Abstract C33E-1617*

Mason, M., **H.-P. Marshall**, Trujillo, E., Marks, D., Glenn, N., and Hedrick, A. (2019). Investigating consistency in snow distribution using a six-year airborne lidar time series of snow depth in Tuolumne Basin, CA, USA. *Fall AGU, San Francisco, CA, Abstract C33E-1623*

Gongora, J., **H.-P. Marshall**, Hase, C., Bradberry, T., and Ballerini, S. (2019). Applications of deep learning and domain adaptation to the study of mountain snow. *Fall AGU, San Francisco, CA, Abstract H33L-2110*

2018 DeAngelo, J., Osterberg, E., Lewis, G., Ferris, D., Hawley, R., Marshall, H., Birkel, S., and Graeter, K. (2018). Surface melt changes and climate forcing in the western greenland percolation zone. *Fall AGU, Washington, D.C., Abstract C43E-1842*

Forster, R., Deeb, E., Marshall, H., and Lund, J. (2018). Insar coherence variability under changing snow conditions at l-band. *Fall AGU, Washington, D.C., Abstract G41B-0700*

Marks, D., Havens, S., Skiles, M., Dozier, J., Bormann, K., Johnson, M., Robertson, M., Hedrick, A., Marshall, H., and Painter, T. (2018). High resolution spatial measurement of snow properties in a mountain environment: A multi-instrument snow properties assessment experiment. *Fall AGU, Washington, D.C., Abstract C13G-1213*

Mason, M. and Marshall, H. (2018). Comparison of snow variability and distribution from in-situ observations and remote sensing: towards optimal sampling strategies for snow hydrology. *Fall AGU, Washington, D.C., Abstract C13D-1174*

Overly, T., Hawley, R., Osterberg, E., Medley, B., Studinger, M., Meehan, T., Kerby, J., Lewis, G., Marshall, H., and McCarthy, F. (2018). Greenland ice sheet surface roughness from kite aerial photography and structure-from-motion photogrammetry. *Fall AGU, Washington, D.C., Abstract C43D-1824*

Rodriguez, C., Marshall, H., and Rodriguez, P. (2018). Application of a k-band fmcw radar for the temporal tracking of dry snow stratigraphy, snowpack melt-freeze states, and mixed snow-rain precipitation events. *Fall AGU, Washington, D.C., Abstract C13D-1179*

2017 Lewis, G., Osterberg, E. C., Hawley, R. L., Marshall, H.-P., Birkel, S. D., Meehan, T. G., Graeter, K., Overly, T. B., and McCarthy, F. (2017b). Spatial variability of accumulation across the western greenland ice sheet percolation zone from ground-penetrating-radar and shallow firn cores. *Fall AGU, New Orleans, LA, Abstract C53B-1038*

Meehan, T., Marshall, H.-P., Bradford, J., Hawley, R. L., Osterberg, E. C., McCarthy, F., Lewis, G., and Graeter, K. (2017). Continuous estimates of surface density and annual snow accumulation with multi-channel snow/firn penetrating radar in the percolation zone, Western Greenland Ice Sheet. *Fall AGU, New Orleans, LA, Abstract C53B-1037*

Gongora, J., **H.-P. Marshall**, and Glenn, N. (2017). Merging datasets from nasa snowex to better understand how snow falls and moves. *Fall AGU, New Orleans, LA, Abstract C13E-1431*

2016 Meehan, T., Osterberg, E. C., Lewis, G., Overly, T. B., Hawley, R. L., Bradford, J., and Marshall, H.-P. (2016). Multi-channel ice penetrating radar traverse for estimates of firn density in the percolation zone, western greenland ice sheet. *Fall AGU, San Francisco, CA, Abstract C53A-0693*

Hedrick, A., Marks, D., Havens, S., Kormos, P., **H.-P. Marshall**, Bormann, K., and Painter, T. (2016). Producing a long-term baseline snow depth distribution from 45 snow-on lidar surveys over a large mountain basin. *Fall AGU, San Francisco, CA, Abstract C23B-0743*

Auger, J., Birkel, S., Maasch, K., Schuenemann, K., Mayewski, P., Osterberg, E., Hawley, R., and **H.-P. Marshall** (2016). Atmospheric circulation and West Greenland precipitation. *Fall AGU, San Francisco, CA, Abstract A53B-0286*

Gongora, J. and **H.-P. Marshall** (2016). A new study of mountain snowpack through graph spectral analysis. *Fall AGU, San Francisco, CA, Abstract C23B-0744*

Deeb, E., **H.-P. Marshall**, Painter, T., Marks, D., Hedrick, A., Havens, S., Forster, R., and Siqueira, P. (2016). Comparison of palsar-2 interferometric estimates of snow water equivalent, Airborne Snow Observatory snow depths, and results from a distributed energy balance snow model (iSnobal). *Fall AGU, San Francisco, CA, Abstract C33B-0771*

Lewis, G., Osterberg, E. C., Hawley, R. L., Koffman, B. G., Marshall, H.-P., Birkel, S. D., and Dibb, J. E. (2016). Albedo spatial variability and causes on the Western Greenland Ice Sheet percolation zone. *Fall AGU, San Francisco, CA, Abstract C41E-0717*

Hetrick, H., **H.-P. Marshall**, Bradford, J., McNamara, J., and Eiriksson, D. (2016). Quantifying the role of lateral flow of water in a sloped mountain snowpack: spatiotemporal patterns in soil moisture and snowmelt (invited). *Fall AGU, San Francisco, CA, Abstract C51D-0686*

Thundercloud, Z., Osterberg, E. C., Ferris, D., Graeter, K. A., Lewis, G., Hawley, R., and **H.-P. Marshall** (2016). Spatial variability of climate signatures recorded in an array of shallow firn cores from the Western Greenland percolation zone. *Fall AGU, San Francisco, CA, Abstract PP51A-2283*

2014 Marks, D., **Marshall, H.P.**, Winstral, A., and Hedrick, A. (2014). The sensitivity of mountain snowcovers to temperature, humidity, and phase change in a warming climate. *Fall Meeting, AGU, San Francisco, CA, Abstract C43E-0438*

Hedrick, A., Marks, D., Winstral, A., and **Marshall, H.P.** (2014a). Evaluating mesoscale numerical weather predictions and spatially distributed meteorologic forcing data for developing accurate swe forecasts over large mountain basins. *Fall Meeting, AGU, San Francisco, CA, Abstract C43D-0431*

Robertson, M. and **Marshall, H.P.** (2014). Ground penetrating radar for measuring snow water equivalent in wet snow. *Fall Meeting, AGU, San Francisco, CA, Abstract C43D-0426*

2013 Havens, S., Johnson, J. B., **Marshall, H.P.**, Nicholson, B., and Trisca, G. (2013a). Snow avalanche detection and identification for near real-time application. *Fall AGU, San Francisco, CA, Abstract S23B-2493*

2013 Hetrick, H., **Marshall, H.P.**, Bradford, J. H., and Mead, J. (2013). Ert inversion with the incorporation of surface-based gpr. *Fall Meeting, AGU, San Francisco, CA, Abstract NS33A-1686*

Watson, K. A., **Marshall, H.P.**, McNamara, J., and Flores, A. (2013). Influence of high-resolution land-surface initialization on near surface hydrometeorological variables in a coupled land-atmosphere model. *Fall Meeting, AGU, San Francisco, CA, Abstract H21A-1011*

Tinkham, W. T., Smith, A., **Marshall, H.P.**, Falkowski, M. J., and Winstral, A. (2013b). Quantifying spatial distribution of snow depth errors from lidar using random forests. *Fall Meeting, AGU, San Francisco, CA, Abstract H13J-1497*

Marshall, H.P., Sturm, M., Elder, K., and Yueh, S. (2013). Extrapolating from pits: Quantifying spatial variability of snow for remote sensing calibration/validation (invited). *Fall Meeting, AGU, San Francisco, CA, Abstract C41C-0650*

Dean, J., **Marshall, H.P.**, Rutter, N., and Karlson, A. (2013). Improving nir snow pit stratigraphy observations by introducing a controlled nir light source. *Fall Meeting, AGU, San Francisco, CA, Abstract C41C-0646*

Aguayo, M., **Marshall, H.P.**, McNamara, J. P., Mead, J., and Flores, A. (2013). Reconstruction of layer densities in a multilayer snowpack using a bayesian approach to inverse modeling. *Fall Meeting, AGU, San Francisco, CA, Abstract C41A-0579*

Hedrick, A., **Marshall, H.P.**, Elder, K., and Yueh, S. (2013). Using ground truth-validated, large-scale airborne lidar to assess the accuracy of the snodas hydrological model in complex terrain. *Fall Meeting, AGU, San Francisco, CA, Abstract C41B-0622* van Wijk, K., Otheim, L., **Marshall, H.P.**, Kurbatov, A., and Spaulding, N. (2013). laser ultrasonic characterization of ice cores. *Fall Meeting, AGU, San Francisco, CA, Abstract C13C-0694*

Trisca, G., Robertson, M., **Marshall, H.P.**, Koenig, L., and Comberiate, M. (2013). Grover: An autonomous vehicle for ice sheet research. *Fall Meeting, AGU, San Francisco, CA, Abstract C13C-0691*

2012 Kormos, P. R., McNamara, J. P., Seyfried, M. S., Marks, D., Flores, A., **Marshall, H.P.**, and Williams, C. J. (2012). Critical zone soil properties effects on soil water storage and flux. *Fall AGU, San Francisco, CA, Abstract H53I-1656*

Deeb, E., **Marshall, H.P.**, LeWinter, A., Finnegan, D., Deems, J. S., and Landry, C. (2012). Comparison of coordinated satellite and ground-based x-band radar collections for the retrieval of snow parameters. *Fall AGU, San Francisco, CA, Abstract C33C-0681*

Evans, S., Heilig, A., Kohn, M., and **Marshall, H.P.** (2012). Isotopic dynamics in a seasonal snowpack. *Fall AGU, San Francisco, CA, Abstract C33C-0673*

2011 Hopkins, A. J., Heilig, A., **H.-P. Marshall**, and Flores, A. N. (2011). Potential application of nasa smap radar data to detect the aerial extent of snowpack melt. *Fall AGU, San Francisco, CA, Abstract H23G-1360*

Havens, S. and **H.-P. Marshall** (2011). Comparison of snowmicropenetrometer estimated microstructure and frequency modulated continuous wave radar. *Fall AGU, San Francisco, CA, Abstract C31A-0600*

Lutz, E., Hawley, R. L., **H.-P. Marshall**, Osterberg, E. C., Courville, Z., Overly, T. B., and Wong, G. J. (2011). Thule to summit: Understanding the physical properties of northern greenland near-surface snow and firn. *Fall AGU, San Francisco, CA, Abstract C33A-0624*

Walters, R. D., McNamara, J. P., **H.-P. Marshall**, and Flores, A. N. (2011). A physically-based approach to downscale coarse snow model output for hydrologic modeling at hillslope scales. *Fall AGU, San Francisco, CA, Abstract C33D-0668*

Tinkham, W. T., Hoffman, C. M., Falkowski, M. J., Smith, A. M., Link, T. E., and **H.-P. Marshall** (2011). Spatial accounting for errors in lidar-derived products: Snow volume and snow water equivalent estimation. *Fall AGU, San Francisco, CA, Abstract C33D-0672*

Deeb, E. J., **H.-P. Marshall**, Finnegan, D. C., Deems, J. S., and Landry, C. (2011). Comparison of ground-based lidar and ground-based radar of southwestern colorado snowpack. *Fall AGU, San Francisco, CA, Abstract C33D-0673*

H.-P. Marshall, Deeb, E. J., Gleason, A., Heilig, A., Finnegan, D. C., Deems, J. S., Havens, S., Kormos, P. R., Landry, C., and McCreight, J. L. (2011). Alpine snow distribution from-ground based radar measurements compared with a high resolution digital elevation model from ground-based lidar observations. *Fall AGU, San Francisco, CA, Abstract C33D-0674*

Heilig, A., **H.-P. Marshall**, Mayer, C., and Winstral, A. H. (2011). Spatial variability of SWE in alpine areas - how do variability patterns change with grid designs? *Fall AGU, San Francisco, CA, Abstract C33E-0686*. (Invited)

Eiriksson, D., McNamara, J. P., **H.-P. Marshall**, and Bradford, J. H. (2011). Assessment of the hydrologic significance of lateral water flow in snow. *Fall AGU, San Francisco, CA, Abstract C33E-0688*

Kormos, P. R., McNamara, J. P., Marks, D. G., Flores, A. N., **H.-P. Marshall**, and Boe, E. (2011). Use of plot scale observations to gauge the applicability of physically-based models. *Fall AGU, San Francisco, CA, Abstract H33H-1418*

Bradford, J. H., **H.-P. Marshall**, and Dickins, D. (2011). Modeling frequency dependent gpr wave propagation through sea ice using the reflectivity method and application to oil-spill detection. *Fall AGU, San Francisco, CA, Abstract NS51A-1742*

2010 Rutter, N., **H.-P. Marshall**, Tape, K. D., and Essery., R. (2010). Observations of snowpack properties to evaluate ground-based microwave remote sensing. *Fall Meeting, AGU, San Francisco, CA, Abstract C33A-0513*

Havens, S., **H.-P. Marshall**, Elder, K., Rutter, N., and Tape, K. D. (2011). Snow micro penetrometer for classifying grain types in the alpine and arctic environment. *Fall Meeting, AGU, San Francisco, CA, Abstract C33A-0514*

Tedesco, M., **H.-P. Marshall**, and Steiner, N. (2010). From colorado to greenland: the 2010 ground passive and active snow (gaps) experiment. *Fall AGU, San Francisco, CA, Abstract C33A-0516*

Shallcross, A. T. J. M. A. F. H. M. D. M. N. G. (2010). Estimating basin snow volume using aerial lidar and binary regression trees. *Fall Meeting, AGU, San Francisco, CA, Abstract C33C-0537*. (Invited)

Deems, J. S., Finnegan, D. C., Deeb, E. J., **H.-P. Marshall**, Bryant, A. C., Skiles, S., Landry, C., and Painter, T. H. (2010). A comparison of ground-based LiDAR, contact spectroscopy, FMCW radar, and manual snow pit profiles of a mountain snowpack. *Fall Meeting, AGU, San Francisco, CA, Abstract C33D-0562*

Deeb, E. J., **H.-P. Marshall**, Deems, J., Finnegan, D. C., Painter, T. H., Landry, C., Bryant, A. C., and Skiles, S. (2010). Field collection efforts of snowpack properties in support of remote sensing applications. *Fall Meeting, AGU, San Francisco, CA, Abstract C33D-0564*

Boe, E., McNamara, J. P., and **H.-P. Marshall** (2010). Quantifying snow variability using an inexpensive network of ultrasonic depth sensors. *Fall Meeting, AGU, San Francisco, CA, Abstract H11A-0796*

2009 Steiner, N., Tedesco, M., **H.-P. Marshall**, and Josberger, E. G. (2009). The 2009 Ground Passive and Active Snow (GAPS) Experiment. *EOS Trans. AGU Fall meeting suppl., Abstract C51B-0483*, 90

Anderson, B., McNamara, J. P., and **H.-P. Marshall** (2009). Sources of snow cover variability in complex terrain. *EOS Trans. AGU Fall meeting suppl., Abstract C31D-0461*, 90

H.-P. Marshall, Koh, G., and Sturm, M. (2009b). A tool for rapid snow depth, stratigraphy, and swe measurements: portable microwave radar. *EOS Trans. AGU Fall meeting suppl., Abstract C31D-0465*, 90

Shallcross, A. T., McNamara, J. P., **H.-P. Marshall**, Marks, D. G., Link, T. E., and Winstral, A. H. (2009). Estimating snow volume in mountain catchments using aerial LiDAR. *EOS Trans. AGU Fall meeting suppl., Abstract C31D-0468*, 90

2009 Demuth, M. N., **H.-P. Marshall**, Morris, E. M., Burgess, D. O., and Gray, L. (2009). The variation of polar firn subject to percolation - characterizing processes and glacier mass budget uncertainty using high-resolution instruments. *EOS Trans. AGU Fall meeting suppl., C31E-0470*, 90

2008 Lutz, E., Birkeland, K., and **H.-P. Marshall** (2008). Important factors influencing microstructural estimates derived from the snowmicropen. *EOS Trans. AGU Fall meeting suppl., Abstract C21C-0579*, 89(53)

McCreight, J., Slater, A., **H.-P. Marshall**, and Deems, J. (2008). Non-stationarity and statistical partitioning of snow depth fields: Independent variable controls and application to sampling design and modeling. *EOS Trans. AGU Fall meeting suppl., Abstract H23B-0963*, 89(53)

2007 **H.-P. Marshall**, Gleason, A., Landry, C., and McCreight, J. (2007d). Snow depth and snow water equivalent distribution in a high alpine basin: quantifying the length scales and magnitude of variation in senator beck basin, colorado. *EOS Trans. AGU Fall meeting suppl., Abstract GC41A-0104*, 88(52)

Deeb, E., Forster, R., **H.-P. Marshall**, and Rutter, N. (2007). Ground-based radar measurements of the northern colorado snowpack at clpx- ii. *EOS Trans. AGU Fall meeting suppl., Abstract C23A-0929*, 88(52)

2006 Schneebeli, M., **H.-P. Marshall**, and Johnson, J. (2006). Snow properties considered from a multiscale perspective: new measurement techniques. *EOS Trans. AGU Fall meeting suppl., Abstract C21C-1163*, 87(52). (Invited)

2003 Gleason, J. and **H.-P. Marshall** (2003). Measurements of snow deformation using particle image velocimetry combined with finite element simulations. *EOS Trans. AGU Fall meeting suppl., Abstract C41B-0952*, 84(46)

2002 **H.-P. Marshall** (2002b). Modeling the interface between two layers of snow using finite elements. *EOS Trans. AGU Fall meeting suppl., Abstract C11B-0996*, 83(47)

Waddington, E., Neumann, T., Morse, D., and **H.-P. Marshall** (2002). Spatial accumulation-rate pattern inferred from radar internal layers and point measurements of velocity and accumulation near taylor mouth, victoria land. *EOS Trans. AGU Fall meeting suppl., Abstract C12A-0998*, 83(47)

2001 Harper, J., Bradford, J., **H.-P. Marshall**, Gleason, J., Pfeffer, T., and Humphrey, N. (2001). Snow stratigraphy in an alpine environment: Spatial variability and a comparison of measurement methods. *EOS Trans. AGU Fall meeting suppl., Abstract IP51A-0740*, 82(47)

BSU Undergraduate Research Symposium, Student Presentations

2012 Burch, K., Drake, K., II, L. L., **H.-P. Marshall**, and Planting, A. (2012). Snow penetrometer embedded system design. *BSU Undergraduate Research and Scholarship Conference*

Peterson, A. and **H.-P. Marshall** (2012). Estimates of snow accumulation rates over the greenland ice sheet using airborne radar. *BSU Undergraduate Research and Scholarship Conference*

Robertson, J. and **H.-P. Marshall** (2012). Snow melt dynamics. *BSU Undergraduate Research and Scholarship Conference*

Computer Programming Skills

Windows, DOS, Macintosh, Unix, Linux operating systems

Matlab (Data Aq., Signal Proc., Statistics, PDE, Wavelet, Symbolic Math Toolboxes),

Developed GUI-driven analysis software for FMCW radar acquisition and processing, snowpit data display, and seismic signal detection of ice quakes.

C++, LaTex, Python, Abaqus, IDL, GPSurvey, LabView, Awk, GMT, Pine, Emacs, Vi

Campbell Scientific data logger communications programs

S+, Mathematica, Surfer, IslandDraw, Excel, Access, Visual Basic

data acquisition, data analysis, signal processing, database management, finite element analysis

low power microcontrollers for ADC/DAC and data logging

Skills / Certifications

Avalanche Science - Rec Level 2+ certification (advanced)

Wilderness First Responder (2017-present)

CPR for the Professional rescuer (2017-present)

UNAVCO InSAR training (2018)

First Responder (1998, EMT program, North Seattle C.C.)

Ground Penetrating Radar (monopulse and FMCW, 5 MHz - 18 GHz)

GPS + theodolite survey equipment, SnowMicroPenetrometer

Northwest Avalanche Institute - Level 1 certification (beginning)

Northwest Avalanche Institute - Level 2 certification (advanced)

American Institute for Avalanche Research and Education - Level 1 Instructor Training

Snow survival and crevasse rescue - U.S. Antarctic Program

experience with snowmobile operation and repair

experience with electronic, metal, and wood-working tools

geophysical instrument design and fabrication

Volunteer Activities

Special Olympics basketball coach (02/15-06/15)

ESL Teacher for refugees (08/09-07/10)

Mentor, Big Brothers / Big Sisters of La Plata County (07/2006-08/2008)

Volunteer high school teacher, Old Crow, Yukon Territory, Canada (1999-2000)

Helped install seismic station on the summit of Mt. Rainier (1999)

Logan School of Learning (2-3rd graders), Field Trip leader to Loveland Pass, CO (2003)

Co-taught 3-day English course to indigenous leaders in southern Chile (2003)

Guest lecturer, Nathan Hale High School Physics class (1999)

Notetaker for Disabled Student Services, University of Washington (1996)

Teen Feed (1995)

Affiliations

Member, Snow Radar Algorithm Team, NASA SCLP (2004-present)

International Glaciological Society (IGS)

American Geophysical Union (AGU, Cryophere Representative 2008-2012)

Sigma Xi (Full member)

American Avalanche Association (Research Chair 2005-2011, Professional Member)

American Institute for Avalanche Research and Education (AIARE) Certified Instructor (2003-04)

American Meteorological Society (Full Member)

Interests / Hobbies

Backcountry skiing / snowboarding, mountaineering, rock climbing, basketball, trail running, mountain biking

West African percussion, Flamenco percussion, yoga

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